

**Bloomberg**

# **DATA LICENSE**

**Per Security Product Manual**

February 2008

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## What is Data License Per Security?

### Introduction

The Data License Per Security Product allows a customer to access financial information on a per security basis. Customers can shape a request to meet their information needs and submit it to Bloomberg through a variety of methods. Bloomberg processes these requests and sends back the information. This is a very flexible, effective, and timely way to access Bloomberg data.

Bloomberg data is divided into four field types: Security Master or Fundamental descriptive data, End of Day Prices, Derived data, and Historical Time series data.

### Request Options:

There are two basic types of requests:

**One-shot** This is a one-time request for data. At any time, the customer submits a **request** file, which Bloomberg processes on receipt or at a time specified by the customer.

**Scheduled** The customer submits a **request** file, to be processed by Bloomberg according to client specification. Options include daily, weekdays, weekends, weekly or monthly. The customer has the option of receiving the entire file every day or just the changes between the current file and the previous file (see **DIFFFLAG** option). The request will run at a time specified by the customer.

Request files are composed of the following sections:

**File header.** This section contains information that identifies the customer, the type of request to run, the time to run the request, and other options that are detailed in the **File Header** chapter. Bloomberg may add options to this section during processing.

**List of desired data items** This **Fields** section may or may not be needed in a request file, depending upon the program being run by the client (i.e., data items must be specified in the **getdata** program; data items are predetermined in the **getticks** program).

**List of securities** This section can be filled with a list of individual securities, a wildcard macro or a link to a portfolio or security search on the BLOOMBERG Professional. Bloomberg offers a **wildcard** mechanism that allows the selection of universes of securities, such as all corporate bonds from a specified country or all stocks that trade on a specified exchange. Secondary qualifiers can be used for a more specific security list.

## Where can I load these files?

### Supported Platforms and Databases

Data License output is sent in flat ASCII text files. Clients can load these files into *any* platform that supports ASCII text.

Bloomberg Data License can be implemented on any hardware platform that can execute FTP Transfer and process delimited ASCII files. Bloomberg Data License supports Internet connectivity.

- **Required Hardware**

Bloomberg Data License is not hardware dependent. If the use of a dedicated circuit is desired, however, clients are responsible for and shall pay for all costs of communication, common carrier equipment installation and monthly charges incurred with the service.

- **Required Software for Internet Connection**

### Encryption

DES software is used to encrypt files sent to accounts with an Internet connection. Further information is provided in the next section of this document and the des.readme text file. The des.readme and several des executables for various platforms are available. Bloomberg will provide this software by putting the executable files in the Notices folder of the client's home directory.

The software is currently supported on the following operating systems: Windows 95, Windows 98, Windows 2000, Windows NT, Windows XP, AIX, AIX1, Solaris, Solaris 2.5, Solaris 2.6, OSF/1 (DEC Alpha), DG/UX and HP/UX. If this software is needed on another platform, Bloomberg will provide a file that can compile a version for the respected platform.

### Compression

The UNIX utility gzip is used to compress files. Files loaded into a UNIX platform can be decompressed with the UNIX gzip utility. Files loaded into a Windows platform can be decompressed with the WinZip application. *Bloomberg always compresses the output files of the **gethistory**, **getticks** and **getallticks** programs before delivery.*

## How do I get my data?

### Access, Delivery Options, and Security

There are two methods by which a customer can request data from Bloomberg: Proprietary and FTP.

- **Proprietary Method**

A Data License client with a BLOOMBERG PROFESSIONAL™ service can leverage the existing infrastructure and avoid additional equipment and communication charges by using DLDL or Send File. Information regarding the DLDL function is available on the BLOOMBERG PROFESSIONAL™ Service under the DLDL Help Key <GO>. Send File information is provided later in this document.

- **FTP (TCP/IP) Method**

Each customer will be assigned a restricted login to Bloomberg's FTP servers. The login, combined with a password, will protect access to a customer's home directories on these servers. Customers send data request files and pick up the resulting data output from these directories via FTP (TCP/IP). Two connectivity possibilities are specified below.

**FTP via the Internet** There is no equipment charge or communication charge for Internet delivery. FTP via the Internet requires all data to be encrypted. We use DES software to encrypt files sent to accounts with an Internet connection.

**FTP via dedicated circuit** If the use of a dedicated circuit is desired, client is responsible for and shall pay for all costs of communication, common carrier equipment installation and monthly charges incurred with the service. There is an installation fee and a monthly equipment charge for the maintenance of hardware.

## Security

**Firewall** To prevent unauthorized access by customers into our network, we maintain a firewall between the Bloomberg mainframes and the Bloomberg FTP servers. The customer may use a firewall to prevent the possibility of any unauthorized connections to their LAN through the Bloomberg routers.

**Internet Encryption** In order to ensure secure data transfers over the Internet, Bloomberg uses the 56-bit symmetric Data Encryption Standard (DES) algorithm to decrypt/encrypt the files (See [Appendix A](#) for copyright details). Bloomberg provides files containing this DES software along with an encryption key. The DES software is currently supported on the following operating systems: Windows 95, Windows 98, Windows 2000, Windows NT, Windows XP, AIX, AIX1, Solaris, Solaris 2.5, Solaris 2.6,

SunOS 5.7, OSF/1 (DEC Alpha), DG/UX and HP/UX. **Bloomberg will provide these files by putting them in the Notices folder of the client's home directory.** Users of Windows-based operating systems should use the software named des.exe. DES files appropriate for the other operating systems have the operating system name included in the file name, (i.e., des-solaris). Further information is available in a des.readme file. The extension **.enc** is added to the encrypted file names. Bloomberg Data License Front End software is available to simplify the handling of encrypted/decrypted files. Please see the Bloomberg Data License Front Ends Software User Manual for further details.

## What do I need to get the files?

### What You Need from Bloomberg

#### For FTP

- A home directory will be created on two interchangeable FTP servers.
- A Login and Password will be assigned for the home directories.
- The IP Addresses and Host Names of the FTP servers will be provided (see [Appendix B](#))
- For FTP via dedicated circuit, communication equipment will need to be purchased from and installed by Bloomberg
- If use of the Data License Front Ends Software applications is desired, please contact your Data License Sales Representative. Upon the receipt of a Per Security contract, Bloomberg will provide the software by placing it in the client's FTP Notices subdirectory. Most Data License Front Ends software is made available at no charge.

#### For DLDL

- The BLOOMBERG PROFESSIONAL™ service is needed to utilize the DLDL function.
- Bloomberg will permission this function for an individual's logon upon receipt of a contract.



## Per Security Request Mechanisms

### Request File Mechanisms – FTP and Send File

The two methods of communication for the process described in this document are FTP (TCP/IP) and Send File via the BLOOMBERG PROFESSIONAL™ service.

#### FTP Requests

Each FTP customer will be given a restricted login for two interchangeable FTP servers. For security purposes, customers will not have permission to access directories outside their home directories. Please refer to [Appendix B](#) for Host Names and IP addresses.

An FTP request has a limit of 127 characters per line in the [header section](#) of the request. All other lines have a limit of 255 characters.

Typically, FTP customers put request files in their home directories and then poll the server for the resulting reply files. To prevent connectivity issues, clients should ensure they close their existing FTP session any time they open a new one. *Files that are sent to Bloomberg for processing must have an extension of .req (any case combination) or the request will not be processed.* The request file name (including the four characters used for .req) must not exceed 25 characters.

When a request file has been sent to our backend machine for processing, the file is moved to .copied. For example, when the request *equity-info.req* is sent to the backend machine, it is renamed *equity-info.req.copied*.

The only accepted extensions are .req and .req.enc (see encryption note below). Requests sent in with the .copied extension in the file name will not be processed. Request file names should not contain an equal sign (=).

Clients can use the [REPLYFILENAME](#) option in the header section of the request file to specify the name of their reply file. The replyfile name must not contain an equal sign (=). The reply file name must not end in .req or .req.enc. The reply file will not appear in the customer's home directory until it has *completely* transferred from our backend machine and is ready to be picked up.

Internet customers can send *encrypted* request files. Bloomberg supplies the encryption software and key. (See [Appendix A](#) for copyright details.) The software is currently supported on the following operating systems: Windows 95, Windows 98, Windows 2000, Windows NT, Windows XP, AIX, AIX1, Solaris, Solaris 2.5, Solaris 2.6, OSF/1 (DEC Alpha), DG/UX and HP/UX. Files containing the DES software are available. Bloomberg will provide these files by putting them in the Notices folder of the client's home directory. These files are additionally available in the DLSD <GO> function of the BLOOMBERG Professional™. Users of Windows-based operating systems should use the des.exe software. DES files appropriate for the other operating systems have the operating system name included in the file name, (i.e., des-solaris). Further information is available in a des.readme file.

Once a key is assigned and the proper version of the software is obtained, the command to encrypt is

```
des -E -u -k "xxxxxxx" request.req request.req.enc
```

The key is represented in the above as xxxxxxxx. The double quotes are required and the key is **case-sensitive**. *request.req* is the clear (readable) file and *request.req.enc* is the encrypted request file. Encrypted request files *must* have an extension of **.req.enc** or they will not be recognized as valid requests.

## Errors in FTP Requests

If errors are detected in a request file (the file does not correspond to specifications), the request file is moved to the request filename appended with **.err** and the errors are written to the reply file (as specified by **REPLYFILENAME** in the header of the request file, which is described in the following section). Error checking is done immediately on the server, and if errors are found, the request is not sent to the backend machine.

For example, if a request file called **prices.req** was created to produce **prices.out**, and **prices.req** had errors, the request file would be moved to **prices.req.err**, and a description of the errors would be contained in **prices.out**. Once a request is placed on the server, it is advisable to wait to see that **.copied** has been appended to the request file name to verify that the request has been accepted for processing.

File names cannot contain the following characters:

```
` ~ ! @ # $ % & * ( ) [ ] ,  
{ } : ; ' " > < / ? \ | =
```

The above special characters are not allowed in the **REPLYFILENAME** as well. Spaces are also not allowed in request or reply file names.

## FTP Requests - Internet

In order to ensure secure data transfer over the Internet, reply files are encrypted before being sent to Bloomberg's FTP servers. The 56-bit symmetric Data Encryption Standard (DES) algorithm is used to encrypt the reply files. Bloomberg supplies software for the decryption of the reply files, as well as the encryption key. (See [Appendix A](#) for copyright details.) The software is currently supported on the following operating systems: Windows 95, Windows 98, Windows 2000, Windows NT, Windows XP, AIX, AIX1, Solaris, Solaris 2.5, Solaris 2.6, OSF/1 (DEC Alpha), DG/UX and HP/UX. Files containing the DES software are available. Bloomberg will provide these files by putting them in the Notices folder of the client's home directory. These files are also available under the DLSD <GO> function of the BLOOMBERG PROFESSIONAL™. Users of Windows-based operating systems should use the des.exe software. DES files appropriate for the other operating systems have the operating system name included in the file name, (i.e., des-solaris). Further information is available in a des.readme file.

The command line is

```
des -D -u -k "xxxxxxx" replyfile.encrypted replyfile.clear
```

The key is represented in the above as xxxxxxxx. The double quotes are required and the key is **case-sensitive**. **replyfile.encrypted** is the encrypted reply file and the final decrypted (i.e. clear and readable) file is **replyfile.clear**. If the encryption key contains a back-quote (') (with the exception of international keyboards), single quotes should be used around the key instead of double quotes for proper decryption.

Some DOS applications require a carriage return as an end-of-line marker in files. In order to add this carriage return to the clear file (i.e. before it is encrypted), the header option FILETYPE=PC is required (See **FILETYPE** section). When the client decrypts the file, these carriage returns will be present.

## Send File Requests

Send File is a method for sending request files via the BLOOMBERG PROFESSIONAL™ service. A Send File request has a limit of 80 characters per line.

The name and path of the file containing the requested data (reply file) is specified by the customer in the header of the request file and is returned to the user. If a reply file name is not specified, it will default to **reply.txt**. See note below under Errors in Send File Requests.

**Send File Requests *must* follow the format below:**

IMPORTANT NOTE: The following lines of code in the request file **MUST** be capitalized for the request file to process correctly:

```
START-OF-FILE
SYSTEM=DATA           (This line is only used for Send File requests)
START-OF-FIELDS
END-OF-FIELDS
START -OF-DATA
END-OF-DATA
END-OF-FILE
```

### START-OF-FILE

SYSTEM=DATA (This *must* be on the second line of the request file as shown.)

<File Header> (See **File Header** Section)

<List of Fields> (See **Data Items** Section)

<List of Securities> (See **Securities** Section)

### END-OF-FILE

**SYSTEM=DATA** This *must* appear on the second line of the request file, immediately after START-OF-FILE. This tells the system that the file is a Data License Send File request, and without this line the file will never be processed.

The beginning of the request file should be similar to the following:

```
START-OF-FILE
SYSTEM=DATA
LOGIN=JD0E
USERNUMBER=123456
SN=987654
WS=0
REPLYFILENAME=out.txt
PROGRAMNAME=getdata
LAUNCH=yes
...
END-OF-FILE
```

A customer can locate the numbers needed above by entering IAM <GO> on the BLOOMBERG PROFESSIONAL™ service. The number after **User:** is the **USERNUMBER**. The first part of the number following **S/N:** (before the hyphen) is the **SN**, and the second part of **S/N:** (after the hyphen) is the **WS** number.

The request file will be uploaded through the BLOOMBERG PROFESSIONAL™ terminal; to do this, the proper file upload setting is required:

- Right-click on a Bloomberg screen and select **Terminal Defaults**
- Select the **Setup** tab
- Click the **Edit...** button next to “**Upload Type**”
- In the “Bloomberg Profiles” popup window, highlight “**Data License**” and click the OK button, then click the OK button on the “**Terminal Defaults**” popup.

To upload a request file, right-click on a Bloomberg screen and select **Upload File...**

Select the request file to be sent and click “**Open**”. The Send File request will be transmitted.

*Note: If “Data License” is not listed as an option under the “Bloomberg Profiles” popup window, please contact [Technical Support](#) for assistance.*

Progress from this point may be monitored under the Bloomberg functions **RPT** (Report Menu) and **FTR** (File Transfer). As an output file is generating, its transmission progress is displayed on FTR. If LAUNCH=yes has been included in the file header, the reply file will be automatically opened by a PC application, depending on the file extension; otherwise, the file will be saved in the download directory without opening. The FTR monitor will hold a transferred file for seven days after initial creation. During this period the transferred file will be available for retransmission.

If there are errors in the request file, they will be returned in the reply file, unless there is an error with the serial number. In this case, an error file will *not* be returned, since the system will not know to which terminal the reply file should be sent.

**Additional note:** To specify a download directory (as opposed to the default), right-click on the BLOOMBERG screen and left-click “Terminal Defaults...” Under the “Setup” tab, click the “Edit...” button next to “Download Setup” Select the path where reply files should be saved in the “Save in:” box and click “Save.” Then click “OK”.

## Errors in Send File Requests

If errors are detected in a request file, they are written to the reply file (as specified by REPLYFILENAME option which is described in the File Header Section). This error checking is done immediately and if errors are found, the request is not processed and the reply file is immediately returned.

If no REPLYFILENAME is specified in the request file, then the file is returned as reply.txt in the default directory as specified in the **Download Setup...** option on the BLOOMBERG PROFESSIONAL™ service (see “Additional note” above).

File names containing the following special characters are *not* valid:

` ~ ! @ # \$ % & \* ( ) [ ] ,  
{ } ; ‘ “ > < / ? | =

## Scheduled Requests

In the case of scheduled requests, reply files are transferred, on a daily, weekday, weekend, weekly, or monthly basis, from the backend machine to the FTP server (see the **PROGRAMFLAG** option in File Header Section). In the case of Send File, the reply files are transferred to the download directory of the BLOOMBERG PROFESSIONAL™ through which the request was scheduled. Multiple scheduled requests can be set up. A different scheduled reply file will be generated for each unique reply file name. Files will be generated as scheduled until cancelled by the user.

To edit securities or fields in a scheduled request, the client must send another request file, listing new securities or fields under the same REPLYFILENAME as the original reply file. A request with a different REPLYFILENAME will be treated as a new scheduled file and the original will continue to run without changes.

## Canceling Scheduled Requests

It is possible to cancel a scheduled job by sending a request file with PROGRAMNAME=cancel. Clients must ensure that the cancel file has the exact same REPLYFILENAME and PROGRAMFLAG as the scheduled reply file. For example, to cancel a daily job which returns data in the file corps.out, set

```
REPLYFILENAME=corps.out  
PROGRAMFLAG=daily  
PROGRAMNAME=cancel
```

## Linking to a Bloomberg Terminal

FTP clients whose firms subscribe to the BLOOMBERG PROFESSIONAL™ service have the option of linking their data license account to their BLOOMBERG terminal to take advantage of personal defaults (for example, third party pricing for fixed income, real-time exchange pricing, etc.). This can be hard-coded for a permanent link by contacting our technical support desk, or this can be done on a case-by-case basis by amending the request file header.

To link to a terminal, the following lines need to be added to the file header:

```
USERNUMBER=  
WS=  
SN=
```

A customer can locate the numbers needed above by entering IAM <GO> on the BLOOMBERG PROFESSIONAL™ service. The number after “User” is the **USERNUMBER**. The first part of the number following “S/N:” (before the hyphen) is the **SN** and the second part of “S/N:” (after the hyphen) is the **WS** number.

*Note: If you receive the following return in your output file, please contact your account representative for assistance:*

**# SN and/or user number not in the same firm as dlxxxxxx**

## Software Roll-Out

When a new beta version of the software used to generate the data returned by Data License request programs is available, clients can process with it through use of the header option VERSION=new (See File Header Section). Please note that if this version of the software is requested, the results will not be as reliable as the production version of the software. Beta versions of software are typically moved into production weekly; this is not guaranteed.

## Request Files

This section outlines the basic file format for the different types of requests that can be made. Request files must be printable ASCII characters only; no binary characters or special escape sequences are allowed. DOS files, which contain an additional carriage return as the end-of-line marker, are accepted. Other file formats (e.g. - Word, Excel) are *not* accepted. The default delimiter character used in request files is “|” (UNIX pipe).

Request files **must** follow this format:

|                      |                                           |
|----------------------|-------------------------------------------|
| START-OF-FILE        |                                           |
| <File Header>        | (See <a href="#">File Header</a> Section) |
| <List of Fields>     | (See <a href="#">Data Fields</a> Section) |
| <List of Securities> | (See <a href="#">Securities</a> Section)  |
| END-OF-FILE          |                                           |

**START-OF-FILE** - Tells the system where to begin reading a file. This **must** be the first non-blank in the file.

**END-OF-FILE** - This record closes out the whole file. It **must** be the last record in the file. If it is not in the file, the request will not be recognized as valid and will never be processed.

Blank lines may appear freely within a request file and will always be ignored, regardless of their position within the file after START-OF-FILE. Lines may also be commented out in request files; any line that starts with # will indicate a comment line and be replicated in the reply file.

END-OF-FILE is the last line the system will read during processing; any text following END-OF-FILE will be ignored.

## File Header

The following table contains variables or header options that may appear in the header section of request files. The second column, “Program Names”, lists the program(s), to which the header options apply; if “all programs” is the return, this indicates **getactions**, **getallticks**, **getdata**, **gethistory**, **getticks**, and **scheduled**. The third column gives the default value for each header option. The fourth and fifth columns indicate which variables are required and which are optional for the two types of request file mechanisms.

*The header section of the file can be changed without notice.* Each header option is described in more detail below. Header options must be entered into the request file in upper case. Bloomberg Data License may insert comments into the header section of a replyfile at any time.



| Header Option  | Program name                               | Default                       | FTP <sup>1</sup>      | Send File <sup>1</sup> |
|----------------|--------------------------------------------|-------------------------------|-----------------------|------------------------|
| ACTIONS        | getactions                                 | all                           | optional              | optional               |
| ACTIONS_DATE   | getactions                                 | entry                         | optional              | optional               |
| CLOSINGVALUES  | getdata                                    | no                            | optional <sup>4</sup> | optional <sup>4</sup>  |
| COLUMNHEADER   | getdata                                    | no                            | optional              | optional               |
| COMPRESS       | All programs                               | variable                      | optional              | optional               |
| DATEFORMAT     | getactions, getdata, gethistory            | mmddyyyy                      | optional              | optional               |
| DATERANGE      | All programs (except getdata)              | No default                    | optional              | optional               |
| DELIMITER      | getdata                                    | (UNIX pipe)                   | optional              | optional               |
| DERIVED        | getdata                                    | no                            | optional <sup>4</sup> | optional <sup>4</sup>  |
| DIFFFLAG       | getdata                                    | no                            | optional              | optional               |
| FILETYPE       | All programs                               | UNIX                          | optional              | n/a                    |
| FIRMNAME       | All programs                               | No default                    | required              | optional               |
| HEADER         | All programs (except scheduled)            | yes                           | optional              | optional               |
| HIST_CRNCY     | getdata, gethistory                        | no                            | optional              | optional               |
| HIST_FORMAT    | gethistory                                 | vertical                      | optional              | optional               |
| HIST_PERIOD    | gethistory                                 | Smallest period               | optional              | optional               |
| HIST_OPTION    | gethistory                                 | No default                    | optional              | optional               |
| HISTORICAL     | getdata                                    | no                            | optional <sup>4</sup> | optional <sup>4</sup>  |
| LAUNCH         | All programs                               | no                            | n/a                   | optional               |
| LOGIN          | All programs                               | No default                    | n/a                   | required               |
| OUTPUTFORMAT   | getdata                                    | variable                      | optional              | optional               |
| PORTSECD       | All programs (except scheduled)            | No default                    | optional              | optional               |
| PROGRAMFLAG    | All programs (except scheduled)            | one-shot                      | optional              | optional               |
| PROGRAMNAME    | All programs                               | getdata                       | optional              | optional               |
| PRP            | All programs (except scheduled)            | No default                    | optional              | optional               |
| QUOTECOMPOSITE | getdata                                    | no                            | optional              | optional               |
| REPLYFILENAME  | All programs                               | See note <sup>2</sup>         | optional              | optional               |
| REPORT         | getdata, gethistory, getticks, getallticks | no for FTP, yes for Send File | optional              | optional               |
| RUNDATE        | All programs                               | Today                         | optional              | optional               |
| SECDESLLENGTH  | getdata                                    | 30                            | optional              | optional               |
| SECID          | All programs (except scheduled)            | No default <sup>3</sup>       | optional              | optional               |
| SECDESLLENGTH  | getdata                                    | no                            | optional <sup>4</sup> | optional <sup>4</sup>  |
| SN             | All programs                               | No default                    | optional              | required               |
| SPECIALCHAR    | getdata                                    | decimal                       | optional              | optional               |
| SYSTEM         | All programs                               | No default                    | n/a                   | required               |
| TICKADJUSTDATE | getticks, getallticks                      | no                            | optional              | optional               |
| TICKEXCHLENGTH | getticks, getallticks                      | 1                             | optional              | optional               |
| TICKLOCALTZ    | getticks, getallticks                      | no                            | optional              | optional               |
| TIME           | All programs                               | No default                    | optional              | optional               |
| USERNUMBER     | All programs                               | No default                    | optional              | required               |
| VERSION        | All programs (except scheduled)            | No default                    | optional              | optional               |
| WS             | All programs                               | No default                    | optional              | required               |
| YELLOWKEY      | All programs (except scheduled)            | No default                    | optional              | optional               |

<sup>1</sup> For the request file mechanism (FTP or Send File) that applies, please see Request File Mechanisms Section.

<sup>2</sup> For FTP requests, the default is <request>.out, where <request> is the request file name less the ".req" extension. For Send File requests, the default is reply.txt.

<sup>3</sup> For a list of valid values for SECID, please see File Header Section.

<sup>4</sup> The header options CLOSINGVALUES, DERIVED, SECMASTER or HISTORICAL *must* be selected for getdata requests from their respective categories, otherwise N.S. (Not Subscribed) will be returned instead of data. They need not be selected if data from the category type they represent is not requested.



**ACTIONS** (optional) - This only applies to the getactions program. This can be used as a filter to request only certain actions or categories of actions. For example, it is possible to request a single action type, such as Acquisitions. It is also possible to request a category of actions along with a single action from another category like Distributions and Mergers. To achieve this, users would specify a “pipe” delimited list of mnemonics (e.g., ACTIONS=DISTRIBUTIONS | MERG). For a list of corporate actions, and categories see the Corporate Actions Reference Guide. For information about the getactions program, see the **Getactions** Section later in this manual.

**ACTIONS\_DATE** (optional) - This flag only applies to the getactions program. This option allows customers to request corporate actions based on different dates. It can be set to the following values:

**entry** - The date the corporate action was entered into the Bloomberg database. If DATERANGE is not used, the system will look for actions entered into the database during the previous 24 hours.

**effective** - The effective date of the corporate action. If DATERANGE is not used, this will default to the current day.

**both** - This incorporates both the entry and effective options.

**CLOSINGVALUES** (optional) – This option only applies to the getdata program. It allows customers to request data via fields from the End of Day Pricing category. If SECMASTER=yes is already in the header, this option need not be selected. If neither SECMASTER nor CLOSINGVALUES is elected, a getdata request will return **N.S.** for an End of Day Pricing field.

**COLUMNHEADER** (optional) - This option only applies to the getdata program. The default is no; if set to yes, the output file will return the title of each column of data. For example, if the fields PX\_BID, PX\_MID, and PX\_ASK were requested, the following row would appear above the data:

```
<Identifier>|<Rcode>|<Nfields>|PX_BID|PX_MID|PX_ASK|
```

The above “pipe” delimiter can be changed using the DELIMITER= option.

**COMPRESS** (optional) - This option returns the requested data file compressed by the UNIX tool gzip. Files may be decompressed using gunzip or Winzip. The output files of clients with an internet connection are automatically encrypted. Internet clients using COMPRESS=yes will need to decrypt first and decompress second. **Note: The getticks, getallticksand gethistory program output files will AUTOMATICALLY be compressed - the COMPRESS=no option is not available for these programs.**

**DATEFORMAT** (optional) - This controls the format of dates. DATEFORMAT defaults to mmddyyyy. There are fourteen optional formats available:

|                  |            |                   |            |                 |            |
|------------------|------------|-------------------|------------|-----------------|------------|
| <b>mmddyy</b>    | 04/28/00   | <b>yyddmm</b>     | 0/28/04    | <b>mmyydd</b>   | 04/00/28   |
| <b>dmmyy</b>     | 28/04/00   | <b>yyyyddmm</b>   | 2000/28/04 | <b>mmyyyydd</b> | 04/2000/28 |
| <b>yyyymmdd</b>  | 20000428   | <b>yymmdd</b>     | 00/04/28   | <b>ddyymm</b>   | 28/00/04   |
| <b>mmddyyyy</b>  | 04/28/2000 | <b>yyyy/mm/dd</b> | 2000/04/28 | <b>ddyyyyym</b> | 28/2000/04 |
| <b>dd-mmm-yy</b> | 28-Apr-00  | <b>ddmmyyyy</b>   | 28/04/2000 |                 |            |

**DATERANGE** (optional) - This option allows control over the date range used for securities in the getactions, getallticks, gethistory and getticks programs. The possible uses of this header option are as follows:

- **DATERANGE=r**

where **r** is an integer specifying the **actual** number of days - not number of business days - from the current day. For example, DATERANGE=7

- **DATERANGE=date1|date2**

where **date1** (start date) and **date2** (end date) are in the format yyyymmdd. For example, DATERANGE=19950101|20050101

The default DATERANGE value for getticks and getallticks is 0; data from the current day will be returned.

**DELIMITER** (optional) - This allows the specification of the delimiter that is used. Any single ASCII character may be used except for " (a double quote) and a blank. The default delimiter is "|" (UNIX pipe). This option will be ignored if OUTPUTFORMAT is set to anything other than variable, which is the default. Only the first character after the equal sign is used. For example:

DELIMITER=,                      Commas will be used

DELIMITER=#\$                    # will be used (\$ is ignored)

For samples, please see [Appendix C](#). Note, if a delimiter is specified and it is not a "|" (UNIX pipe), all text fields will be surrounded with double quotes.

**DERIVED** (optional) – This option only applies to the getdata program, and allows customers to request data via fields from the Derived Data category. Without DERIVED=yes in the header, a getdata request for Derived Data fields will return **N.S.**

**DIFFFLAG** (optional) - This controls the Bloomberg output being sent back to the customer. It is applicable to scheduled files (see the **PROGRAMFLAG** variable later in this section) using the getdata program **ONLY**. Bloomberg can send only the changes from the previous reply file. DIFFFLAG defaults to "no" and has the following valid values:

**no** - This means that *all* output (unmodified) will be sent to the customer.

**yes** – The file generated will contain the output of the UNIX "diff" utility. In cases where something has changed, this "diff" will provide full records from the previous reply file and the current reply file. New securities will be apparent, as there will be no previous record. Deleted securities will be apparent, as there will be no current security. The customer can reconstruct the current reply file by combining the previous reply file with the changes that Bloomberg has sent. Bloomberg only recommends this option if users are familiar with the UNIX "diff" utility and are confident they can reconstruct the file.

**changes** - As with the **yes** option, the UNIX diff utility will be used to compare the previous reply file to the current reply file. In this case, the file sent to the customer will contain only additional securities and securities that have received an update. The full record of each security will be sent. Deletes will not be represented.

In output files, the DIFFFLAG line will be returned between TIMESTARTED and START-OF-DATA.

Example request header:

```
START-OF-FILE
RUNDATE=20050518
FIRMNAME=dl123456
FILETYPE=pc
REPLYFILENAME=difftest.out
DIFFFLAG=changes
PROGRAMFLAG=weekday
SECMASTER=yes
PROGRAMNAME=getdata
```

Example output file:

```
...
TIMESTARTED=Wed May 18 11:48:52 EDT 2005
DIFFFLAG=changes
START-OF-DATA
IBM US Equity|0|3|INTL BUSINESS MACHINES CORP|1613.321|.740|
MSFT US Equity|0|3|MICROSOFT CORP|10804.354|3.320|
INTC US Equity|0|3|INTEL CORP|6173.000|.240|
END-OF-DATA
TIMEFINISHED=Wed May 18 11:48:53 EDT 2005
END-OF-FILE
```

**FILETYPE** (optional) - *This option applies to Internet customers only!* If FILETYPE = PC, then a carriage return is added to the end of each line of the reply file before it is encrypted. This allows the decrypted file to be opened in various DOS applications which require the carriage return to signal the end of a line.

**FIRMNAME** (required for FTP only) - This is the login name assigned by Bloomberg. If a client sends an FTP request file with an incorrect FIRMNAME, it will be returned as an error. The FIRMNAME value is **case-sensitive**. For example,

FIRMNAME=dl123456

**HEADER** (optional) - This allows customers to specify if the header should be returned in the reply file. This option can be set to either yes, no or timeonly, where yes is the default.

- If **HEADER=no** is specified, only the data between START-OF-DATA and END-OF-DATA is returned; if **HEADER=no** is used in conjunction with **COLUMNHEADER=yes**, the first line of the output file will be

SECURITIES|ERROR CODE|NUM FLDS|

- If **HEADER=timeonly** is specified, the TIMESTARTED and TIMEFINISHED lines will be included in the output file.

**HIST\_CRNCY** (optional) – This option allows for the specification of a desired currency for history requests. It applies only to historical data and can be used in both the getdata and gethistory programs. It is available for use in the getdata program only with the **single-point history fields** listed in the Getdata Section. A currency ISO code must be specified. The Bloomberg sub-currency (GBP, ZAR, etc.) codes are not available for use since they are not true ISO codes. This option is not applicable to all securities in all cases.

**HIST\_FORMAT** (optional) – This option will alter the output format of a gethistory request. The default **vertical** format separates the data by field requested and can be considered vertical in appearance. In the default format each field becomes a header over its related dates and values. The **horizontal** format combines the output of all fields into a single record per date for each security. For examples, please see the **gethistory** section.

**HIST\_PERIOD** (optional) - Sets the periodicity of gethistory requests. The default behavior is to provide the smallest period available for the data requested. Possible valid values are **d** (daily), **w** (weekly, Friday), **m** (monthly, last day of month), **q** (quarterly, last day of the quarter), **y** (yearly, last day of the year).

**HIST\_OPTION** (optional) - This allows users of the gethistory program to retrieve historical averages for the date range and period specified. The only valid value is "average." HIST\_PERIOD must be set to weekly or greater.

**HISTORICAL** (optional) - This option is for the getdata program only, and needs to be set to YES (HISTORICAL=yes) for Historical Time Series fields; otherwise, **N.S.** will be returned. The following are Historical Time Series fields:

|                            |                     |
|----------------------------|---------------------|
| MTG_HIST_WAC               | MTG_HIST_COLLAT_BAL |
| MTG_HIST_DLQ30             | MTG_HIST_CPN        |
| MTG_HIST_DLQ60             | EQY_DVD_HIST        |
| MTG_HIST_DLQ90             | EQY_DVD_HIST_ALL    |
| MTG_HIST_FORECL            | EQY_DVD_HIST_SPLITS |
| MTG_HIST_REO               | EQY_DVD_HIST_GROSS  |
| MTG_HIST_WAM               | EQY_DVD_ADJUST_FACT |
| MTG_HIST_ORIG_AMT          | EQY_DVD_ADJ_FUND    |
| MTG_HIST_REM_AMT           | MTG_HIST_FACT       |
| MTG_HIST_NUM_LOANS         | MTG_HIST_PREPAY     |
| MTG_HIST_PREPAY_FRCST_BMED | FACTOR_SCHEDULE     |

**LAUNCH** (only an option for Send File) - This value may be set to **yes** or **no**; **no** is the default. If it is set to yes, upon transfer of the reply file, an attempt will be made to open the reply file using the appropriate application. The reply filename extension determines which application will be launched; this association is usually handled automatically by the operating system or can be set by the customer on his/her own PC.

**LOGIN** (required by Send File only) - This is the login name of the user on the BLOOMBERG PROFESSIONAL™ service. For example,

LOGIN=JSMITH

**OUTPUTFORMAT** (optional) - This option controls the format of output files created with the getdata program. Three specifications exist:

**bulklist** - This option applies to bulk fields only (See [Getdata Section](#)). When this is specified, each entry in the bulk field is listed, one per line. This option will work if *only bulk fields* are requested; if bulk and non-bulk fields are requested within the same file, the option will be ignored.

**fixed** - This option separates each column without using a delimiter. It is based solely on spacing (see the section called [Data Items \(Fields\)](#) for more information on specifying field widths). See [Appendix C](#) for examples.

**variable** - This is the default. Field information is returned separated by the specified delimiter (or the default UNIX pipe). Each field is returned with the delimiters separating them without any extra spaces before or after the data. See [Appendix C](#) for examples

**PORTSECDES** (optional) - This option may be used to specify the way in which the security description is returned in the reply file when a PORTFOLIO macro is used. By default, the data in the field SECURITY\_DES is returned in the first column, but when this option is set to **adjusted**, the identifier is returned as it appears on the Bloomberg in the portfolio.

**PROGRAMFLAG** (optional) - This flag determines how often to process the request. There are six values currently available. If this option is not specified, it will default to **one-shot**.

**one-shot** - This specifies that the request is to be run one time and one time only. The request is serviced immediately, unless otherwise specified (see **TIME** and **RUNDATE** variables).

**daily** - This specifies that the request is to be processed daily at a certain time (see **TIME** variable below).

**weekly** - This specifies that the request is to be processed weekly, on the same day of the week the initial request was submitted. Time of day can be selected using the **TIME** variable.

**monthly** - This specifies that the request is to be processed monthly, on the same day of the month the initial request was submitted. The day of the month can be selected using the **RUNDATE** variable, and time of day can be specified using the **TIME** variable.

**weekday** - This specifies that the request is to be processed on Monday through Friday only. This option can be used in conjunction with **TIME** and **RUNDATE** variables, and is valid for all programs. It looks at an account's region (New York, London or Tokyo) in order to base the day on the applicable time zone.

**weekend** - This specifies that the request is to be processed on Saturday and Sunday only. This option can be used in conjunction with **TIME** and **RUNDATE** variables and is valid for all programs. It looks at an account's region in order to base the day on that time zone.

If the **TIME** variable is not used in a scheduled request, the file will immediately process at the time the request is received by Bloomberg, and will thereafter run at **0000** (midnight) of the local region.

**PROGRAMNAME** (optional) - This names the program to be run by Bloomberg. The following are available programs:

**getdata** - This program retrieves various data fields from the database(s) for the specified list of securities. See the **Getdata** Section for further information.

**gethistory** - This program retrieves various historical data fields from the database(s) for the specified list of securities within the given date range. See the **Gethistory** Section for further information, including the list of fields available for this program.

**getticks** - This program retrieves price and volume trade ticks for the specified list of securities within the given date range. A maximum of 50 business days of ticks is available. See **Getticks** Section.

**getallticks** - This program retrieves price and volume ticks (including matched bids and asks) for the specified list of securities within the given date range. A maximum of 50 business days of ticks is available. See [Getallticks](#) Section.

**getactions** - This program retrieves one day's worth of corporate actions for a specified list of securities (and the issuer of these securities). See [Getactions](#) Section.

**cancel** - This program cancels a currently running scheduled request file. To cancel, send a second request with REPLYFILENAME equal to that of the scheduled reply file. For example, to cancel a daily job that returns data in the file corps.out, set

```
REPLYFILENAME=corps.out  
PROGRAMFLAG=daily  
PROGRAMNAME=cancel
```

**scheduled** - This program returns a report containing all request files that are currently scheduled. Only the required Header Options noted in the File Header Section and the START-OF-FILE and END-OF-FILE line are needed in a request file with PROGRAMNAME set to this value. This is always treated as a one-shot request.

A request file will look like this:

```
START-OF-FILE  
FIRMNAME=d1123456  
REPLYFILENAME=scheduled.out  
PROGRAMNAME=scheduled`  
END-OF-FILE
```

If the PROGRAMNAME option is not specified, it will default to getdata. See Data Dictionary Section for information regarding a list of fields that are applicable to the above programs.

**PRP** (optional) – This is the PRP Setup number. The PRP Setup number can be found on the BLOOMBERG PROFESSIONAL™ service. Enter **PRP <GO>** and choose the number of the applicable report, found on the left of the Portfolio Report Table under the column header “RUN”. This option can be used with or without the PORTFOLIO Macro type described in the Wildcards section. Clients also have the option of requesting additional fields between START-OF-FIELDS and END-OF-FIELDS. The output file will return the additional requested fields, along with those from the PRP setup.

**QUOTECOMPOSITE** (optional) - This option is for the getdata program only, and needs to be set to YES (QUOTECOMPOSITE=yes) for the Bond Quote Composite field (BOND\_QUOTE\_COMP) to populate.



**REPLYFILENAME** (optional) - This specifies the name of the file that will be created to contain the output of a data request. The filename can be any string not exceeding 12 characters for Send File and 25 characters for FTP. It should not contain any blanks or any of the following characters:

‘ ~ ! @ # \$ % & \* ( ) [ ]  
{ } ; ‘ “ > < / ? | =

Additionally, the colon (":") and backslash ("\") are not allowed for FTP requests, but are allowed for Send File requests. If a reply file name is the same as a previous one, the old file is overwritten. Examples:

REPLYFILENAME=equity.pricing.job.1  
REPLYFILENAME=corporates.dump

If not specified, REPLYFILENAME defaults to the request filename with **.out** as the extension in place of **.req** for FTP, and it defaults to **reply.txt** for Send File. For example, a request file called **test.req** that contains no REPLYFILENAME will return **test.out** for FTP requests; the same request file run via Send File will return **test.txt**.

If a reply file's name contains the above-listed invalid characters, or ends in either **.req** or **.req.enc**, the request file will be moved to the request filename appended with **.err**

**REPORT** (optional for FTP, standard for Send File) – The possible values are *no*, *yes*, and *only*.

**no** – This is the default for FTP requests; no report will be sent.

**yes** – This is standard for Send File and optional for FTP. A report table is created and returned alongside the replyfile. The report provides a count of the valid securities returned in the replyfile. For FTP this table will be provided as a separate file with **.rpt** as the extension. For Send File this table is listed under the **RPT <GO>** function of the BLOOMBERG PROFESSIONAL™.

**only** – Only the report table with the security counts is returned; no replyfile will be generated.

**RUNDATE** (optional) - This option controls the date on which requests are processed. For requests scheduled with a frequency made available under the PROGRAMFLAG option, RUNDATE specifies the date on which the job will begin. The date cannot be more than seven days in the future.

The syntax is **RUNDATE=YYYYMMDD**, where YYYY is the year including century, MM is the month and DD is the day. For example, to start a request on June 15, 2005, set **RUNDATE=20050615**

This option can be used in conjunction with the **TIME** option. If no TIME is specified, the file will start at 00:00 of the date specified, or immediately if the date is current day.

In every reply file, RUNDATE is returned in the header as the second line, below START-OF-FILE.



**SECDESLength** (optional) - This option allows specification of the length of the security description in the reply file. The security description is the first data column that is returned and valid values for this number are 1 through 32. For example:

|                  |                                                         |
|------------------|---------------------------------------------------------|
| SECDESLength=8   | Security description of 8 characters                    |
| SECDESLength=100 | This will default to the maximum width of 32 characters |

For sample files, please see [Appendix C](#).

**SECID** (optional) - This option allows the specification of a default security identifier (see [Data Items section](#)). If an alternate security identifier is specified for a security, this option is ignored for that security. For example, if the bulk of the securities are being requested by ISIN number, SECID=ISIN may be stated in the header and only the ISIN numbers need to be specified in the data section for each record. In the example below, TICKER and SEDOL are assigned as alternate identifiers; the remaining numbers will process as ISIN:

```
SECID=ISIN
...
START-OF-DATA
US4592001014
US8855351040
CT30 Govt |TICKER|
2346070 |SEDOL|
DE0005245500
JP3592200004
...
END-OF-DATA
```

The security identifier must be one of the following:

|                              |           |            |
|------------------------------|-----------|------------|
| AUSTRIAN                     | CZECH     | JAPAN      |
| BB_UNIQUE                    | DUTCH     | LUXEMBOURG |
| BELGIAN                      | EUROCLEAR | SEDOL      |
| CATS                         | FRENCH    | SPAIN      |
| CEDEL                        | IRISH     | TICKER     |
| CINS                         | ISIN      | VALOREN    |
| COMMON_NUMBER                | ISRAELI   | CUSIP      |
| WPK (Wertpapier Kenn-Nummer) | ITALY     |            |

If the wrong SECID is provided, the output file will contain return code 10 (for “security not found”). In the above example, if an ISIN was incorrectly identified as a sedol -

```
JP3592200004 | SEDOL
```

the resulting output file would return

```
JP3592200004 | 10 | ...
```

If requesting equity securities by their CUSIP identifiers, it must be specified that the identifier being used is a CUSIP. By way of example, the CUSIP for IBM's common stock is 459200101. In a request file, one of the following formats must be used:

459200101|CUSIP  
459200101 Equity|CUSIP

in order for the security to be recognized for processing.

When requesting by TICKER, the market sector must be specified; see **YELLOWKEY** section below.

**SN** and **WS** (required by Send File, optional for FTP) - These are the serial and workstation numbers of the BLOOMBERG PROFESSIONAL™ service. This information can be found by entering IAM <GO> on the terminal. There are two numbers separated by a "-" after "S/N:" The first number (preceding the hyphen) is the serial number. The second number (following the hyphen) is the workstation number. These options are required for Send File requests; they are available to FTP customers to link their terminal permissions (e.g., real-time exchange pricing) to an FTP login on a per request basis.

**SECMaster** (optional) – This option only applies to the getdata program. It allows customers to request data via fields from both the End of Day Pricing and the Security Master categories. Without SECMaster selected, a request for Security Master data will return **N.S.** End of Day Pricing should be requested independently with CLOSINGVALUES=yes.

**SPECIALCHAR** (optional) - This flag controls the output of fractional and decimal fields. Fractional characters can occur in price fields and security descriptions. SPECIALCHAR defaults to decimal and can currently take three values:

**bloomberg** - Use Bloomberg's special characters where appropriate, e.g., one half is represented as a byte containing 0x9f (See Getdata Section).

**fraction** - Translate Bloomberg's special characters to ASCII fractions, e.g., one half is represented as 1/2.

**decimal:n** - Translate Bloomberg's special characters to ASCII decimals; e.g., one half is represented as .5. The number of decimal places can be specified using the format decimal:n, where n is the number of decimal places desired. When the number of decimal places is specified, the number is rounded; e.g., for SPECIALCHAR=decimal:3, the number 98.6175 will be returned as 98.618. If no :n is specified, the number of decimal places defaults to 6, which is the maximum precision.

Certain fields return security descriptions, such as SECURITY\_DES for corporate bonds. Security descriptions always come back with "fraction characters":

IBM 4 1/8 06/30/05

**SYSTEM** (required by Send File, not applicable for FTP) The only value is DATA. SYSTEM=DATA must appear as the second line of a Send File request, immediately after START-OF-FILE. This specifies that the request is a Data License Send File request; without this line, the file will never be processed.

**TICKADJUSTDATE** (optional) - This header option is available for getticks and getallticks; the default is YES. This option is used to specify that the date returned for each line of tick date should be adjusted to be based on New York time. This only occurs for those exchanges that trade across midnight NY time (Asia region exchanges). When TICKADJUSTDATE is set to NO, the date of the tick data will be the close date. Below is an example of a New Zealand security, with the file processed on a New York machine:

TICKADJUSTDATE=yes

```
START-OF-DATA
START SECURITY|KIP NZ Equity|
KIP NZ Equity|05/06|01:34:30|T|1.15|A||CL|
KIP NZ Equity|05/06|01:34:30|B|1.14|A||A|1.15|A|||
KIP NZ Equity|05/06|00:47:20|T|1.15|A|91|SP|
KIP NZ Equity|05/06|00:46:49|T|1.15|A|105||
KIP NZ Equity|05/06|00:22:03|T|1.15|A|4909|SP|
KIP NZ Equity|05/06|00:17:33|T|1.15|A|17000|SP|
KIP NZ Equity|05/06|00:06:44|T|1.15|A|810||
KIP NZ Equity|05/05|23:10:38|T|1.15|A|23333|SP|
KIP NZ Equity|05/05|23:08:48|T|1.15|A|7153||
...
```

TICKADJUSTDATE=no

```
START SECURITY|KIP NZ Equity|
KIP NZ Equity|05/06|01:34:30|T|1.15|A||CL|
KIP NZ Equity|05/06|01:34:30|B|1.14|A||A|1.15|A|||
KIP NZ Equity|05/06|00:47:20|T|1.15|A|91|SP|
KIP NZ Equity|05/06|00:46:49|T|1.15|A|105||
KIP NZ Equity|05/06|00:22:03|T|1.15|A|4909|SP|
KIP NZ Equity|05/06|00:17:33|T|1.15|A|17000|SP|
KIP NZ Equity|05/06|00:06:44|T|1.15|A|810||
KIP NZ Equity|05/06|23:10:38|T|1.15|A|23333|SP|
KIP NZ Equity|05/06|23:08:48|T|1.15|A|7153||
...
```

**TICKEXCHLENGTH** (optional) - This option is available for getticks and getallticks programs. Using TICKEXCHLENGTH=2 will return a two character exchange code instead of the default, which is a single character exchange code. These two character exchange codes are the same codes returned by equities in the EXCH\_CODE field. They are available in two lookup tables - LU\_EQY\_PRIM\_EXCH and LU\_COMPOSITE\_EXCH\_CODE.

**TICKLOCALTZ** – This option is available for the getticks and getallticks programs. The default for this option is no, and the default time zone is New York. With TICKLOCALTZ=yes in a request, the time zone setting of the client profile linked to the request will determine the time and date returned in the output. **TZDF <GO>** is the function on the BLOOMBERG Professional™ that controls time zone settings. On a request basis clients can link in their TZDF profiles via the USERNAME option. On an account basis clients can contact [technical support](#) and request that a particular UUID be associated with their account. This option will only work with linked request files.

**TIME** (optional) - This flag determines the time at which request files are processed. The time will be set according to the local time of the account's sales region - Tokyo, London or New York. If TIME is not specified in a request, it is processed immediately. The format is TIME=HHMM, where HH is the hour (00 - 23) and MM is the minute (00 - 59). For example, to start a request at 8:35 PM local time, set TIME=2035; to start a one-shot request at 3:00 PM, set TIME=1500.

**USERNAME** (required by Send File, optional for FTP) - This is the user number of the BLOOMBERG PROFESSIONAL™ service login. To find the USERNAME, type IAM <GO> on the terminal; the user number is located after USER: in the upper left-hand side of the screen. This is required for Send File requests, but it also allows FTP customers to link their personal Bloomberg terminal defaults (e.g., fixed income pricing sources) to an FTP login on a per request basis.

**VERSION** (optional) - If **new** is specified, the new beta version of the software will be used to process the request. As explained in Request File Mechanisms Section, the new beta version of the software is typically moved in weekly. There may be no changes seen for individual Per Security programs. Please note that if this version of the software is requested, the results will not be as reliable as the production version of the software.

**WS** (required by send File, optional for FTP) - Please see the [SN and WS](#) option description above.

**YELLOWKEY** (optional) - This option specifies a default market sector (i.e. Govt, Corp, Equity, etc.) that will be used with the security descriptions. If a market sector description is already appended to the security in the data section, the given value for this option will be ignored. For example,

```
YELLOWKEY=Equity
...
START-OF-DATA
IBM US||
CT30 Govt||
MSFT US||
END-OF-DATA
```

In the above, "IBM US" will be treated as "IBM US Equity" and "MSFT US" will be treated as "MSFT US Equity." "CT30 Govt" will be treated as requested.

Market sector must be one of the standard Bloomberg market sectors:

|       |        |        |
|-------|--------|--------|
| Govt  | Muni   | Comdty |
| Corp  | Pfd    | Index  |
| Mtge  | Equity | Curncy |
| M-Mkt |        |        |

***Market sectors are case-sensitive and must be entered as above.***

## Data Items (Fields)

Fields must be listed between the following two declarations:

### START-OF-FIELDS

<Field Mnemonics>

### END-OF-FIELDS

The list of fields must be after the header and before the list of securities. A maximum of 500 fields can be specified in any request file.

Example:

```
START-OF-FIELDS
ID_ISIN
NAME
PX_ASK
MTG_WAC
EQY_BETA
YLD_CONV_ASK
END-OF-FIELDS
```

Additionally, the way in which field information is returned may be specified (all are optional):

<field mnemonic>| <field width> | <decimal places> | <FP format>

**<field width>** is a number from 1 to a maximum of 30000. It represents the size (number of characters) of the field that is returned. If a value is not specified, the defaults are as follows:

- If the OUTPUTFORMAT is specified as **variable** (the default), the resulting field will be returned surrounded by | (UNIX pipes).
- If the OUTPUTFORMAT is **fixed**, the size of the field defaults to a predetermined fixed size for that field. (The predetermined sizes of fields are represented in the Standard Width column of the *fields.csv* table that can be found in all FTP account directories. Please see Data Dictionary Section for a description of this table).

**<decimal places>** is a number from 0 to 9 and it allows specification of the number of decimal places that are returned with numerical fields. For character fields, this value is ignored. If this option is not specified, the number of decimal places will vary according to field. Columns in the *fields.csv* table provide information about each fields' decimal places. Please see Data Dictionary Section for a description of the *fields.csv* table.

**<FP format>** is a special representation of real numbers without a decimal point (floating point format). To turn this feature on, place a **c** or **C** as the third option after the field mnemonic. If this option is not specified, this feature is turned off by default.

Examples of FP format:

| Normal return | # decimal places | Field width | FP format return |
|---------------|------------------|-------------|------------------|
| 9.7342        | 6                | 10          | 6xx9734200       |
| 14.24353      | 3                | 14          | 3xxxxxxxx14244   |
| 4.234         | 4                | 6           | 442340           |
| 1.4           | 0                | 5           | 0xxx1            |
| 0.906         | 9                | 10          | 9906000000       |

Note: |'s (UNIX pipes) in the FP format return (column 4 above) show the boundary of the field and x's are shown to represent spaces. For example request and reply files using these special formatting features, please see [Appendix C](#).

## Securities

Securities must be listed between the following two declarations:

### START-OF-DATA

<Securities>

### END-OF-DATA

Any text inserted between END-OF-DATA and END-OF-FILE will not be returned in the output. Bloomberg Data License may insert comments in the area between END-OF-DATA and END-OF-FILE at any time.

The list of securities must be after the list of fields. There may only be one security identifier per line, and a maximum of 20,000 securities may be requested in a single file. The following formats can be used:

<Identifier> <Market Sector>  
<Identifier> <Market Sector> | <Security Identifier> |  
<Identifier> | <Security Identifier>

<Identifier>: required.

<Market Sector>: This is optional, but recommended. Processing will be faster if a market sector is specified since Bloomberg will know which security database to search first. Market sector must be one of the standard Bloomberg market sectors specified previously in File Header Section under the **YELLOWKEY** header description.

<Security Identifier> must be one of the identifiers specified previously under the **SECID** header description.

It is also possible to identify securities using security descriptions such as Ticker/Coupon/Maturity or Ticker/Exchange; in these cases, the market sector must be specified.

Some security identifiers apply to multiple securities from multiple exchanges. It is possible to specify the security desired by adding exchange code.

Examples:

```
START-OF-DATA
# Intel Corp
US4581401001 US|ISIN|
US4581401001 US Equity|ISIN|
INTC US Equity
# T14 11/15/11
000863149|VALOREN|
000863149 Govt|VALOREN|
T14 11/15/11 Govt
END-OF-DATA
```



## Wildcards

This section describes wildcards (or macros), which represent a group or universe of securities. All currently available wildcards for the BLOOMBERG Data License product are discussed here, as well as their usage. The general syntax for wildcard usage in a request file is:

*macrotype = primary qualifier [AND secondary qualifier = value [AND secondary qualifier = value AND... ]]*

Wildcards are placed in the data section of a request file. Secondary qualifiers are optional.

## Macrotypes

| Macrotype   | Description           | Primary Qualifier                                               | Notes                                                                                                                                                                                                                        |
|-------------|-----------------------|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BOND_SRCH   | Bond search           | Custom criteria set number                                      | Bond search as set up under the SRCH function on the BLOOMBERG PROFESSIONAL™ service. This includes Corp and Govt. The <b>LOGIN</b> and <b>USERNUMBER</b> header options are required.                                       |
| EQUITY_SRCH | Equity search         | Custom criteria set number                                      | Equity search as set up under the QSRC function on the BLOOMBERG PROFESSIONAL™ service. The <b>LOGIN</b> and <b>USERNUMBER</b> header options are required.                                                                  |
| EXCH        | Equity exchanges      | Two character exchange code, e.g., UN (New York Stock Exchange) | See <b>Appendix D</b> for information about the Lookup Table that returns a list of exchanges and composite exchanges.                                                                                                       |
| INDEX       | Equity indices        | Index ticker symbol, e.g., INDU (Dow Jones Industrial Average)  | See <b>Appendix D</b> for examples of index ticker symbols.                                                                                                                                                                  |
| PFD_SRCH    | Preferred Bond Search | Custom criteria set number                                      | Preferred Bond search as set up under the PSCH function on the BLOOMBERG PROFESSIONAL™ service. The <b>LOGIN</b> and <b>USERNUMBER</b> header options are required.                                                          |
| PORTFOLIO   | Your portfolio        | The Port # of your portfolio                                    | The <b>LOGIN</b> and <b>USERNUMBER</b> header options are required if your Data License account is not linked to a BLOOMBERG PROFESSIONAL™ service. The <b>PRP</b> header option can be used in conjunction with this macro. |
| SECTYP      | Security type         | Security type descriptor, e.g. CORPORATES                       | See the table below.                                                                                                                                                                                                         |

The above bond search wildcards enable customers to download a specific search that has been set up via the BLOOMBERG function **SRCH**, which searches all corporate and government bonds, or **PSCH**, which searches all preferred bonds. All that needs to be specified is the Custom Criteria Set of interest. For example, if a client would like to download SRCH criteria set number 3, this would be specified as

BOND\_SRCH = 3 | MACRO

in the data section of the request file. All bonds that match the given criteria will be returned with the fields that were requested.

The above **PORTFOLIO** wildcard enables customers to utilize an existing portfolio's security set. The line format is the same as the bond search. For example,

PORTFOLIO = 1523 | MACRO

would be used to download the securities of a portfolio whose Port # was 1523. A Port # can be found on the BLOOMBERG PROFESSIONAL™ service. Enter **PRTU <GO>**; the Port #'s are listed to the far right of the "Your Portfolios" table. The PRP header option can be used with the PORTFOLIO macro. For example, PRP=1 will return fields in PRP Setup #1. See File Header Section for further details.

## Security Types

The following table lists all available primary qualifiers (security type descriptors) for SECTYP.

| Security Type           | Description                                                                        | Market Sector |
|-------------------------|------------------------------------------------------------------------------------|---------------|
| ABS                     | Asset-backed securities                                                            | Mtge          |
| BASIS_SW_CRNCY          | Basis swap rates                                                                   | Curncy        |
| CD_CRNCY                | CD rates                                                                           | Curncy        |
| CMBS                    | Commercial Mortgage Backed Securities                                              | Mtge          |
| CMO                     | Collateralized Mortgage Obligations                                                | Mtge          |
| CONVERTS                | Convertible bonds <sup>†</sup>                                                     | Corp          |
| CONVERTS_PFD            | Convertible preferred bonds <sup>†</sup>                                           | Pfd           |
| CONVERTS_UNDERLYING     | Underlying securities to convertible bonds <sup>†</sup>                            | Equity        |
| CONVERTS_PFD_UNDERLYING | Underlying securities to convertible preferred bonds <sup>†</sup>                  | Equity        |
| CORPORATES              | Corporate bonds (includes convertibles) <sup>†</sup>                               | Corp          |
| CORP_PFD                | Corporate preferred bonds                                                          | Corp          |
| CORP_PFD_DOMESTIC       | Corporate and preferred bonds issued by the U.S. or Canada <sup>†</sup>            | Corp, Pfd     |
| CORP_PFD_INTERNATIONAL  | Corporate and preferred bonds <i>not</i> issued by the U.S. or Canada <sup>†</sup> | Corp, Pfd     |
| CORP_WARRANT            | Corporate Warrants                                                                 | Corp          |
| CPS_FLRS_CRNCY          | Caps/floors rates                                                                  | Curncy        |
| CROSS_CRNCY             | Cross exchange rates                                                               | Curncy        |
| DELIVERABLE_BONDS       | Deliverable bonds on futures contracts <sup>†</sup>                                | Comdty        |
| DEPOSIT_CRNCY           | Deposit exchange rates                                                             | Curncy        |
| EQUITY_INDEX            | Equity Indices                                                                     | Index         |
| EURODOLLAR_BOND         | Eurodollar bonds (USD bonds deposited in Europe banks) <sup>†</sup>                | Corp          |

| Security Type   | Description                                                                                            | Market Sector         |
|-----------------|--------------------------------------------------------------------------------------------------------|-----------------------|
| FHLMC_GOLD_POOL | Federal Home Loan Mortgage Corporation securities ("Freddie Mac" gold)                                 | Mtge                  |
| FHLMC_POOL      | Federal Home Loan Mortgage Corporation securities ("Freddie Mac")                                      | Mtge                  |
| FNMA_POOL       | Federal National Mortgage Association securities ("Fannie Mae")                                        | Mtge                  |
| FRA_CRNCY       | Forward Rate Agreements                                                                                | Curncy                |
| FUT_CHAIN       | Futures contracts <sup>†</sup>                                                                         | Comdty                |
| GNMAI_POOL      | Government National Mortgage Association securities ("Ginnie Mae" I)                                   | Mtge                  |
| GNMAII_POOL     | Government National Mortgage Association securities ("Ginnie Mae" II)                                  | Mtge                  |
| GOVT_AGENCY     | U.S. and foreign government agencies <sup>†</sup>                                                      | Corp                  |
| GOVT_NATIONAL   | Non-U.S. government bonds (National) <sup>†</sup>                                                      | Corp                  |
| GOVT_REGIONAL   | Non-U.S. government bonds (Regional) <sup>†</sup>                                                      | Corp                  |
| LOAN            | Corporate Loans                                                                                        | Corp                  |
| NDF_CRNCY       | Non-deliverable forward rates                                                                          | Curncy                |
| NDF_SW_CRNCY    | Non-deliverable forward swaps                                                                          | Curncy                |
| ONSH_CRNCY      | Onshore forward rates                                                                                  | Curncy                |
| ONSH_SW_CRNCY   | Onshore forward swap rates                                                                             | Curncy                |
| OPT_CHAIN       | Equity, future and index option chain <sup>‡</sup>                                                     | Comdty, Equity, Index |
| OPT_VOL_CRNCY   | Option volatility rates                                                                                | Curncy                |
| PFD_BOND        | Preferred bonds (no convertibles) <sup>†</sup>                                                         | Pfd                   |
| QUARTER_CRNCY   | Quarterly swap rates                                                                                   | Curncy                |
| REPO_CRNCY      | Repo rates                                                                                             | Curncy                |
| SEMI_CRNCY      | Semi-annual swap rates                                                                                 | Curncy                |
| SPOT_CRNCY      | Spot exchange rates                                                                                    | Curncy                |
| SUPRA_NATIONAL  | Inter-American Development Bank, Supranational & World Bank bonds (includes convertibles) <sup>†</sup> | Corp                  |
| SWAP_CRNCY      | Swap currency rates                                                                                    | Curncy                |
| SWAP_SPR_CRNCY  | Swap spread rates                                                                                      | Curncy                |
| SWAP_VOL_CRNCY  | Swaption volatility rates                                                                              | Curncy                |
| US_TREASURY     | U.S. treasury bonds <sup>†</sup>                                                                       | Govt                  |
| WHEN_ISSUED     | When Issued securities                                                                                 | Corp                  |
| WHOLE_LOAN      | Whole loan mortgages                                                                                   | Mtge                  |
| X_FWD_CRNCY     | Cross forward rates                                                                                    | Curncy                |

<sup>†</sup> All matured bonds are weeded out of the list of securities returned.

<sup>‡</sup> Must be used in conjunction with the secondary qualifier SECURITY\_DES. See examples below.

Wildcards are placed between "START-OF-DATA" and "END-OF-DATA" in the request file. The usage of wildcards in a request file is best illustrated through the following examples:

```

START-OF-DATA
INDEX=INDU | MACRO
EXCH= UN | MACRO
SECTYP=CONVERTS | MACRO
SECTYP=GOVT_REGIONAL | MACRO
SECTYP=US_TREASURY | MACRO
END-OF-DATA

```

The first example given above will return all members of the Dow Jones Industrials (ticker = INDU). The second example will retrieve all listings on the New York Stock Exchange. The third example will generate the list of all convertible bonds in the corporate database, and so on.

Wildcards can also be used to get history data, tick data, or corporate actions. The following data request returns daily high prices for members of the S&P 500 Index from June 15, 2002 through June 15, 2003

```
START-OF-FIELDS
PX_HIGH
END-OF-FIELDS
```

```
START-OF-DATA
INDEX = SPX | MACRO | 20020615 | 20030615 |
END-OF-DATA
```

In general, wildcards should be treated as any other data request with the addition of “| MACRO” after the universe of securities of interest. Wildcards are meant to facilitate downloading large groups of related securities and security list maintenance. They are *not* designed to function as a general search engine with highly specific search constraints.

## Secondary Qualifiers

Secondary qualifiers can be used to limit or constrain wildcards to a subset of securities. They must be preceded by **AND**. The following table lists all secondary qualifiers and to which macrotypes/primary qualifiers these apply.

| Secondary Qualifier | Description                              | Macrotypes/Primary Qualifiers                                                                                                                                          | Notes                                                                                                                                               |
|---------------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| ACTIVE_PRICING      | Actively priced exchange traded equities | EXCH                                                                                                                                                                   | Valid values are yes and no, where yes is the default. If yes, only those equities priced in the last 30 days are returned; if no all are returned. |
| COUNTRY             | Country (CDR code)                       | CONVERTS, CONVERTS_PFD, CONVERTS_PFD_UNDERLYING, CONVERTS_UNDERLYING, CORPORATES, GOVT_AGENCY, GOVT_NATIONAL, GOVT_REGIONAL, EURODOLLAR_BOND, PFD_BOND, SUPRA_NATIONAL | See <a href="#">Appendix D</a> for information about the Lookup Table that returns a list of valid CDR codes for COUNTRY.                           |

| Secondary Qualifier | Description                     | Macrotypes/Primary Qualifiers                                                                                                                                                                                                  | Notes                                                                                                                                                                       |
|---------------------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CPN_TYP             | Coupon type                     | CONVERTS, CONVERTS_PFD, CONVERTS_PFD_UNDERLYING, CONVERTS_UNDERLYING, CORP_PFD_DOMESTIC, CORP_PFD_INTERNATIONAL, CORPORATES, GOVT_AGENCY, GOVT_NATIONAL, GOVT_REGIONAL, EURODOLLAR_BOND, PFD_BOND, SUPRA_NATIONAL, US_TREASURY | The only valid value for CPN_TYP is FLOATING, which specifies floating rate bonds.                                                                                          |
| CRNCY               | Currency (ISO code)             | CONVERTS, CONVERTS_PFD, CONVERTS_PFD_UNDERLYING, CONVERTS_UNDERLYING, CORP_PFD_DOMESTIC, CORP_PFD_INTERNATIONAL, CORPORATES, GOVT_AGENCY, GOVT_NATIONAL, GOVT_REGIONAL, PFD_BOND, SUPRA_NATIONAL                               | See <a href="#">Appendix D</a> for information about the Lookup Table that returns a list of valid ISO codes for CRNCY.                                                     |
| EXCH_TRADED         | Exchange traded preferred bonds | CONVERTS, CONVERTS_PFD, CONVERTS_PFD_UNDERLYING, CONVERTS_UNDERLYING, PFD_BOND                                                                                                                                                 | The valid values are yes and no. The default is no.                                                                                                                         |
| MATURED             | Matured Securities              | CONVERTS, CONVERTS_PFD, CORPORATES, CORP_PFD_DOMESTIC, CORP_PFD_INTERNATIONAL, EURODOLLAR_BOND, GOVT_AGENCY, GOVT_NATIONAL, GOVT_REGIONAL, PFD_BOND, SUPRA_NATIONAL, US_TREASURY                                               | The valid values are <b>yes</b> and <b>no</b> . The default is <b>no</b> .                                                                                                  |
| NET_CPN             | Net coupon for mortgage pools   | FHLMC_GOLD_POOL, FHLMC_POOL, FNMA_POOL, GNMAI_POOL, GNMAII_POOL                                                                                                                                                                | Any numeric value. A range of values is acceptable.                                                                                                                         |
| RATE                | Mortgage rate                   | FHLMC_GOLD_POOL, FHLMC_POOL, FNMA_POOL, GNMAI_POOL, GNMAII_POOL                                                                                                                                                                | Valid values are ADJUSTABLE or FIXED.                                                                                                                                       |
| RATING              | Bloomberg composite rating      | CONVERTS, CONVERTS_PFD, CONVERTS_PFD_UNDERLYING, CONVERTS_UNDERLYING, CORP_PFD_DOMESTIC, CORP_PFD_INTERNATIONAL, CORPORATES, GOVT_AGENCY, GOVT_NATIONAL, GOVT_REGIONAL, EURODOLLAR_BOND, PFD_BOND, SUPRA_NATIONAL              | See <a href="#">Appendix D</a> for information about the Lookup Table that returns a list of valid Bloomberg composite ratings for RATING. A range of values is acceptable. |
| SECURITY_DES        | Security description            | DELIVERABLE_BONDS, FUT_CHAIN, OPT_CHAIN                                                                                                                                                                                        | Security description as recognized by Bloomberg. See Data Items Section.                                                                                                    |
| SECURITY_TYP        | Security type                   | EXCH                                                                                                                                                                                                                           | See a table below for a list of valid security types                                                                                                                        |

| Secondary Qualifier | Description                                  | Macrotypes/Primary Qualifiers                                                                                                                                                                                                  | Notes                                                                            |
|---------------------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| SERIES              | Series of corporate bond                     | CONVERTS, CONVERTS_PFD, CONVERTS_PFD_UNDERLYING, CONVERTS_UNDERLYING, CORP_PFD_DOMESTIC, CORP_PFD_INTERNATIONAL, CORPORATES, GOVT_AGENCY, GOVT_NATIONAL, GOVT_REGIONAL, EURODOLLAR_BOND, PFD_BOND, SUPRA_NATIONAL              | Any valid series is acceptable, such as 144A.                                    |
| TICKER              | Ticker symbol                                | CONVERTS, CONVERTS_PFD, CONVERTS_PFD_UNDERLYING, CONVERTS_UNDERLYING, CORP_PFD_DOMESTIC, CORP_PFD_INTERNATIONAL, CORPORATES, GOVT_AGENCY, GOVT_NATIONAL, GOVT_REGIONAL, EURODOLLAR_BOND, PFD_BOND, SUPRA_NATIONAL, US_TREASURY | Any valid ticker symbol is acceptable.                                           |
| WAC                 | Weighted average coupon for mortgage pools   | FHLMC_GOLD_POOL, FHLMC_POOL, FNMA_POOL, GNMAI_POOL, GNMAII_POOL                                                                                                                                                                | Any numeric value. A range of values is acceptable.                              |
| WALA                | Weighted average loan age for mortgage pools | FHLMC_GOLD_POOL, FHLMC_POOL, FNMA_POOL, GNMAI_POOL, GNMAII_POOL                                                                                                                                                                | Any numeric value. A range of values is acceptable. This is in number of months. |
| WAM                 | Weighted average maturity for mortgage pools | FHLMC_GOLD_POOL, FHLMC_POOL, FNMA_POOL, GNMAI_POOL, GNMAII_POOL                                                                                                                                                                | Any numeric value. A range of values is acceptable. This is in number of months. |

Currently available security types for the secondary qualifier SECURITY\_TYP are listed below.

| SECURITY_TYP        | Description       |
|---------------------|-------------------|
| AMERICATRUST        | America's trust   |
| CLOSEENDFUND        | Closed end fund   |
| COMMON <sup>1</sup> | Common stock      |
| MONEYMARKET         | Money market fund |
| MUTUAL              | Mutual fund       |
| OFFSHOREFUND        | Off-shore fund    |
| OPTION              | Option            |
| RECEIPT             | Receipt           |
| RIGHT               | Right             |
| UKTRUST             | UK unit trust     |
| UNIT                | Unit trust        |
| WARRANT             | Warrant           |

<sup>1</sup> SECURITY\_TYP=COMMON will return the securities that were listed on an exchange as of the previous day; for example, if a security is added to an exchange on Monday, it will not be returned under this macro until Tuesday.

Secondary qualifiers use the operators =, != (not equal to), > and <. Some examples of secondary qualifiers are:

- `SECTYP=CORPORATES AND CRNCY=USD AND COUNTRY=US|MACRO`  
All corporate bonds issued by the U.S. in U.S. dollars
- `SECTYP=GOVT_REGIONAL AND COUNTRY!=CA AND CRNCY=FRF|MACRO`  
All regional government bonds, not issued by Canada, denominated in French francs.
- `SECTYP=GOVT_NATIONAL AND COUNTRY=CH|MACRO`  
All government national bonds issued by China
- `SECTYP=GOVT_AGENCY AND RATING=AAA AND CRNCY=BE|MACRO`  
All government agency bonds, issued by Belgium, rated AAA
- `SECTYP=US_TREASURY AND TICKER=T|MACRO`  
All U.S. treasuries with "T" as the ticker symbol
- `EXCH=UN AND SECURITY_TYP=COMMON|MACRO`  
All common stocks listed on the New York Stock Exchange.

The deliverable bonds, futures contracts and option chain macros require the secondary qualifier `SECURITY_DES`. For example,

```
SECTYP=DELIVERABLE_BONDS AND SECURITY_DES=USA Comdty|MACRO
SECTYP=OPT_CHAIN AND SECURITY_DES=IBM US Equity|MACRO
```

The first macro returns all deliverable bonds for the US long bond future and the second returns all options for IBM's US listing.

Ranges can be specified for the secondary qualifiers `NET_CPN`, `RATING`, `WAC`, `WALA` and `WAM` by using > and < symbols, as in the following examples:

```
SECTYP=GOVT_AGENCY AND RATING>B2 AND RATING<A3|MACRO
SECTYP=CORPORATES AND RATING<AAA AND RATING>B1|MACRO
SECTYP=CONVERTS AND RATING<AA1|MACRO
SECTYP=GNMAI_POOL AND NET_CPN>6.9 AND NET_CPN<8.0|MACRO
SECTYP=FHLMC_POOL AND WAC>7.25|MACRO
```

The first example retrieves government agency bonds that are rated higher than B2 and lower than A3. The second example gets corporate bonds that are rated lower than AAA and higher than B1. The next example is for convertible bonds that are lower than AA1, etc.

In general, wildcards are easy to use and expedite the generation of lists of securities of a given group. Some basic syntax rules for wildcards follow.

- Wildcards must begin with a macrotype (e.g., INDEX) followed by an equal sign (=). Only one macrotype is allowed per line.
- The order of secondary qualifiers (e.g., CRNCY) does not matter.
- Secondary qualifiers must be preceded by **AND**.
- Only one type of secondary qualifier can be specified per line (except RATING and Mtge qualifiers). For example, COUNTRY = CA AND COUNTRY = CH is not allowed, nor is CRNCY != USD AND CRNCY != FRF.
- Either a range of RATINGS can be specified or one RATING can be asked for, but not both. That is, either use RATING > ??? AND RATING < ??? or RATING = ???, but not RATING = ??? AND RATING > ???. This applies to the mortgage secondary qualifiers NET\_CPN, WAC, WALA and WAM as well.
- A maximum of seven secondary qualifying statements (e.g., TICKER = T) are allowed per line. Any secondary qualifying statement beyond the seventh AND is ignored.

Possible return codes for wildcards are (in general, any non-zero return code means an error has occurred):

|     |                                                                    |
|-----|--------------------------------------------------------------------|
| 0   | Good return. No errors occurred.                                   |
| 500 | Invalid wildcard (macro).                                          |
| 501 | Unknown macrotype.                                                 |
| 502 | Unknown secondary qualifier.                                       |
| 503 | Unknown EXCH.                                                      |
| 504 | Unknown INDEX.                                                     |
| 505 | Unknown SECTYP.                                                    |
| 506 | Unknown CRNCY.                                                     |
| 507 | Internal database error.                                           |
| 508 | Unknown COUNTRY.                                                   |
| 509 | Internal database error.                                           |
| 510 | Invalid operator, i.e. operator other than =, !=, > or < was used. |
| 511 | Invalid RATING request                                             |
| 512 | Multiple secondary qualifiers of the same type.                    |
| 513 | Invalid CPN_TYP.                                                   |
| 514 | Invalid Mtge request.                                              |
| 515 | Invalid Mtge RATE request.                                         |
| 516 | Internal database error.                                           |
| 517 | Options not available.                                             |
| 518 | Deliverable bonds not available.                                   |
| 519 | Futures not available.                                             |
| 520 | Bond search error.                                                 |
| 521 | Portfolio error.                                                   |



## Reply Files (Returned Data)

This section outlines the basic reply file formats for the different programs that can be requested via PROGRAMNAME. Reply files are printable ASCII characters only, with no binary characters or special escape sequences. (Exceptions are described below). The default delimiter character used in reply files is “|” (UNIX pipe) unless otherwise specified. Reply files have the following general format:

**START-OF-FILE** - This will always be the first line in the file.

<Header section>

<Field section>

**TIMESTARTED**=date and time

**DIFFFLAG**=<diffflag> (scheduled “diff” requests only)

<Data section>

**TIMEFINISHED**=date and time

**END-OF-FILE** - This will always be the last record in the file.

Both the header and field sections are exactly as specified in the request file sent by the customer. Lines that were commented in the request file using a # character are simply replicated in the reply file.

**TIMESTARTED** - This is the date and time the request began processing on Bloomberg's backend server. Its format is the same as the output of the UNIX date command with no arguments. For example, a file that started processing on May 19, 2005 at 11:43:56 EDT would return

TIMESTARTED=Thu May 19 11:43:56 EDT 2005

**DIFFFLAG** - This is only valid for the reply files of scheduled “diff” requests. Further information about the **DIFFFLAG** variable can be found in File Header Section. For more information about scheduling requests, please see the **PROGRAMFLAG** variable in the File Header Section.

**TIMEFINISHED** - This is the date and time the request finished processing on Bloomberg's backend server. The format is the same as TIMESTARTED.

TIMEFINISHED=Thu May 19 11:46:51 EDT 2005

## Historical Price Corrections for Per Security Accounts

Historical Price Correction files list securities that have undergone a historical pricing correction. The corrections will have been entered during the previous day. The files are named with the clients account number followed by the extension **.hpc** (i.e., 12345.hpc, if 12345 were a client account). The files are put into clients FTP directories on an as-needed basis.

Only securities that the client has downloaded within the past two months will be included in the file. If the securities that have received corrections are not found in a client's download list, no **.hpc** file will be sent.

Historical price correction records are formatted as follows:

<security description> | <field mnemonic> | <old value> | <old date entered> | <new value> | <new date entered> | <time entered> | <exchange indicator> |

The following fields are being tracked for price corrections:

|                     |                          |
|---------------------|--------------------------|
| BOOK_VAL_PER_SH     | INDX_PX_SALES            |
| CG_SALES            | INDX_UNCH_VOL            |
| CNVX_OAS_BID        | IS_EPS                   |
| CNVX_OAS_MID        | IS_SPECIAL_EPS           |
| DUR_ADJ_ASK         | MF_BLK_1D                |
| DUR_ADJ_BID         | MF_NONBLCK_1D            |
| DUR_ADJ_MID         | MKT_VAL_OF_EQY           |
| DUR_ADJ_OAS_ASK     | MTG_PREPAY_TYP           |
| DUR_ADJ_OAS_BID     | MTG_WAL                  |
| DUR_ASK             | NET_INCOME               |
| DUR_BID             | OPEN_INTEREST            |
| DUR_MID             | OPEN_INTEREST_TOTAL_CALL |
| EQY_DVD_HIST        | OPEN_INTEREST_TOTAL_PUT  |
| EQY_DVD_HIST_GROSS  | PX_ASK                   |
| EQY_DVD_SH_LAST     | PX_BID                   |
| EQY_WEIGHTED_AVG_PX | PX_FIXING                |
| HIST_CALL_IMP_VOL   | PX_HIGH                  |
| HIST_PUT_IMP_VOL    | PX_LAST                  |
| INDX_ADJ_PE         | PX_LOW                   |
| INDX_ADJ_POS_PE     | PX_OPEN                  |
| INDX_ADJ_POS_PX_EE  | PX_VOLUME                |
| INDX_ADJ_PX_EE      | SPREAD_BA_CR             |
| INDX_ADV_VOL        | TOTAL_VOLUME_EQY         |
| INDX_DECL_VOL       | VOLUME_TOTAL_CALL        |
| INDX_GENERAL_EST_PE | VOLUME_TOTAL_PUT         |
| INDX_MKT_VAL        | YLD_CONV_ASK             |
| INDX_POS_ERN        | YLD_CONV_BID             |
| INDX_POS_EST_ERN    | YLD_CONV_MID             |
| INDX_PX_BOOK        |                          |

## Overrides

A distinguishing feature of the BLOOMBERG Data License getdata product is the ability to perform calculation overrides to return personalized analytic values. This allows clients to submit data that can leverage BB standard models and calculation servers.

The value of an overriding field is altered to affect the return of another field. There is a limit of nine overrides per record. Fields that are not affected by the overriding field will return their normal value.

WARNING: If the override is used improperly, incorrect or unexpected data may be returned, or the file may not process at all.

### Request File Format:

<Identifier> || <Number of Overrides> | <Overriding Field> | <Overriding Value> |

Example: **XS0142898286 Corp||1|SETTLE\_DT|20030615|**

<Identifier>|<Security Identifier>|<Number of Overrides>|<Overriding Field>|<Overriding Value>|

Example: **XS0142898286|ISIN|1|SETTLE\_DT|20030615|**

Sample Request File for an override in the getdata Program

```
START-OF-FILE
FIRMNAME=firmabc
REPLYFILENAME=override.out
PROGRAMNAME=getdata
PROGRAMFLAG=one-shot
DERIVED=yes
```

```
START-OF-FIELDS
MTG_PREPAY_TYP
MTG_PREPAY_SPEED
MTG_WAL
END-OF-FIELDS
```

```
START-OF-DATA
# security with no overrides
073914VW0 Mtge
# security with overrides
073914VW0 Mtge||2|MTG_PREPAY_TYP|MTG_PREPAY_SPEED|CPR|90|
END-OF-DATA
END-OF-FILE
```

Sample Reply File for an override in the getdata Program

```
START-OF-FILE
RUNDATE=20050526
PROGRAMFLAG=oneshot
REPLYFILENAME=842.out
PROGRAMNAME=getdata
PROGRAMFLAG=one-shot
DERIVED=yes

START-OF-FIELDS
MTG_PREPAY_TYP
MTG_PREPAY_SPEED
MTG_WAL
END-OF-FIELDS

TIMESTARTED=Thu May 26 08:42:48 EDT 2005
START-OF-DATA
# security with no overrides
073914VW0 Mtge|0|3|CPR|15|3.35|
# security with overrides
073914VW0 Mtge|0|3|CPR|90|.34|
END-OF-DATA
TIMEFINISHED=Thu May 26 08:42:49 EDT 2005
END-OF-FILE
```

Clients who link their files to a Bloomberg terminal can use a pricing source override to download prices from a subscribed fixed income pricing source. The override field is PRICING\_SOURCE. The below asks for Zions Bank's (ZNBK) price:

```
912828DM9 Govt|1|PRICING_SOURCE|ZNBK|
```

The PRICING\_SOURCE override will work only with fixed income securities (Pfd, Govt or Corp) within the **getdata** program.

Clients can use a similar method for overriding currency pricing sources. The default currency pricing source for an unlinked per-security account is Bloomberg Composite New York. Rather than overriding the PRICING\_SOURCE field, the proper format for overriding a currency pricing provider is

```
<TICKER> <Currency source code> Curncy
```

For example, for clients who subscribe to WM/Reuters pricing on the Bloomberg terminal (provider code WMCO), the following would be used to download WM/Reuters' price for the Euro spot rate in a linked per-security request file

```
EUR WMCO Curncy
```

Currency pricing provider codes can be found on the Bloomberg terminal under the **XDF <go>** function.

## **Bloomberg Front End Software**

### **Data License Request Builder and FTP Client Application**

The Data License Request Builder is a tool designed to help Data License clients create BLOOMBERG Data License request files. Bloomberg will provide the software by putting the executable files into the Notices folder of the client's home directory.

This is a fast and convenient method that can be used in lieu of writing request files. A significant feature of this software is the Field Finder, which contains the list of all available fields and their descriptive information. This is the equivalent of the *fields.csv* file, also known as the Data Dictionary.

The Data License FTP Client is a tool designed to help the Data License clients transfer files to and from the ftp servers. The FTP Client is built into the Request Builder in order to allow customers to FTP their files with a few simple steps. It is available to all customers with a Data License Account.

(Note: If the Send File option is used in this application, the FTP Utilities will not be available.)

Please contact [Data License Technical Support](#) with any issues about these applications.

New versions of the software are released as upgrades are made. New versions will replace the old versions when they are posted to the Notices folder of FTP directory.

For clients with the Bloomberg PROFESSIONAL™ Service, all above applications, documentation, and enhancement notices can be downloaded via DLSD<GO>. Help pages are available for all above applications; please also see the Front Ends Software user manual.

### **Data License Transformer Application**

The BLOOMBERG® Data License Transformer is a tool designed to help BLOOMBERG® Data License clients perform custom transformations to standard data files. Its capabilities include the ability to map data to custom data for specified fields, map field names to custom field names, and transform the file format based on selected criteria. Please speak to your account representative for further information on this application.

The Data Mapper section is designed to select custom data to be returned in place of the actual data content returned by Bloomberg in a particular field. The Field Mapper section is designed to select custom field names to be returned in place of the actual Bloomberg field name.

## Per Security Product Information

### File Commands/Programs

Data can be obtained by writing simple commands/programs. There are 5 different programs:

- Getdata
- Gethistory
- Getticks
- Getallticks
- Getactions

The below sections describe how each one is used.

### Getdata

The getdata program returns requested data items on a given list of securities. Data items fall in one of four field categories: **security master** (e.g., coupon, ticker, call schedule), **derived data** (e.g., yield to maturity, effective duration), **historical time series** and **end of day pricing**. One record is returned per security identifier. Fields are pipe-delimited and of variable width. The getdata program can be used in conjunction with wildcards. For information regarding a list of available data items, see the “Data Dictionary” section.

Format: <Identifier> | <Rcode> | <Nfields> | <Value-1> | <Value-2> | ... | <Value-Nfields> |

|                             |                                              |
|-----------------------------|----------------------------------------------|
| <Identifier>                | Security identifier used in the request file |
| <Rcode >                    | Return code                                  |
| <Nfields>                   | Number of fields requested and received      |
| <Value-1>...<Value-Nfields> | Data element #1...Data element # Nfields     |

The following **return codes** are currently defined:

|     |                                                           |
|-----|-----------------------------------------------------------|
| 0   | Good return. No errors occurred.                          |
| 10  | Bloomberg cannot find the security as specified.          |
| 988 | System Error on security level                            |
| 989 | Unrecognized pricing source                               |
| 990 | System Error (Contact <a href="#">Technical Support</a> ) |
| 991 | Invalid override value (e.g., bad date or number)         |
| 992 | Unknown override field                                    |
| 993 | Maximum number of overrides (10) exceeded                 |
| 994 | Permission denied.                                        |
| 995 | Maximum number of fields exceeded.                        |
| 996 | Buffer Overflow (some data for this security is missing). |
| 997 | General override error (e.g., formatting error)           |
| 998 | Security identifier type (e.g., CUSIP) is not recognized. |
| 999 | System error (Contact <a href="#">Technical Support</a> ) |

In addition, it is possible to get a good return code, but have one or more fields where data is not returned. The possible reasons for this are:

- The field is not applicable, such as asking for equity shares outstanding on a U.S. Treasury Bond. Not applicable fields will come back blank.
- The data is missing because Bloomberg does not have the data. When data is not available, the field will return **N.A.**
- The user does not have permission to download the field (e.g., information provided by a 3rd party, or field is not supported for data license). In this case, the field will return **N.D.** (Not Downloadable).
- The user has no contractual agreement to download a certain type of field for a certain security type. In this case, the field will return **N.S.** (Not Subscribed).
- The user has not flagged the proper field category in the request file header (SECMaster, CLOSINGVALUE, DERIVED, HISTORICAL); in this case, the field will return **N.S.** (Not Subscribed)
- Bloomberg does not recognize the field. In this case, the returned value is **FLD UNKNOWN**

### **Applicable getdata file header options**

|                      |                       |                     |
|----------------------|-----------------------|---------------------|
| <b>CLOSINGVALUES</b> | <b>HISTORICAL</b>     | <b>SECDESLength</b> |
| <b>COLUMNHEADER</b>  | <b>LAUNCH</b>         | <b>SECID</b>        |
| <b>COMPRESS</b>      | <b>OUTPUTFORMAT</b>   | <b>SECMaster</b>    |
| <b>DATEFORMAT</b>    | <b>PORTSECDES</b>     | <b>SN</b>           |
| <b>DELIMITER</b>     | <b>PROGRAMFLAG</b>    | <b>SPECIALCHAR</b>  |
| <b>DERIVED</b>       | <b>PROGRAMNAME</b>    | <b>SYSTEM</b>       |
| <b>DIFFFLAG</b>      | <b>PRP</b>            | <b>TIME</b>         |
| <b>FILETYPE</b>      | <b>QUOTEComposite</b> | <b>USERNUMBER</b>   |
| <b>FIRMNAME</b>      | <b>REPLYFILENAME</b>  | <b>VERSION</b>      |
| <b>HEADER</b>        | <b>REPORT</b>         | <b>WS</b>           |
| <b>HIST_CRNCY</b>    | <b>RUNDATE</b>        | <b>YELLOWKEY</b>    |

## **Bulk format**

“Bulk format” is a self-defining data type used to encapsulate multi-dimensional (matrix) data such as call schedules, put schedules, option chains, prepayment vectors, index members, etc. The bulk format is a large character buffer that contains all the elements in the matrix and extra control information.

The format for bulk data is as follows:

1st character: Delimiter used within this bulk format field  
Number of dimensions  
Delimiter  
Number of rows  
Delimiter  
Number of columns  
Delimiter  
...Data Elements...  
Delimiter

The “Data Elements” are a series of delimited values preceded by their data type. For example, the pair

1/15/02, 103.21

will be described as

5;01/15/2002;3;103.2100;

where “5” indicates a date, “01/15/2002” is the actual date, “3” indicates a price and “103.2100” is the actual price. A semicolon is the only allowable delimiter in this format. See the [Bulk Field Data table](#) for a list of data types and data element numbers.

### **Example:**

US00206QAS30 Corp has the following call schedule:

| DATE:     | PRICE:  |
|-----------|---------|
| -----     | -----   |
| 5/15/2005 | 104.06  |
| 5/15/2006 | 103.66  |
| 5/15/2007 | 103 1/4 |
| 5/15/2008 | 102.84  |
| 5/15/2009 | 102.44  |
| 5/15/2010 | 102.03  |
| 5/15/2011 | 101.63  |
| 5/15/2012 | 101.22  |
| 5/15/2013 | 100.81  |
| 5/15/2014 | 100.41  |
| 5/15/2015 | 100     |



This would be described in bulk format as follows (full schedule not displayed, and new line is for readability only):

```
|;2;11;2;5;05/15/2005;3;104.0600;5;05/15/2006;3;103.6600;5;05/15/2007...  
5;05/15/2014;3;100.4100;5;05/15/2015;3;100.0000;|
```

This is translated as:

|            |                         |
|------------|-------------------------|
| ;          | Delimiter               |
| 2          | Number of dimensions    |
| 11         | Number of rows          |
| 2          | Number of columns       |
| 5          | Next element is a date  |
| 05/15/2005 | Date                    |
| 3          | Next element is a price |
| 104.0600   | Price                   |
| ...        |                         |
| 5          | Next element is a date  |
| 05/15/2015 | Date                    |
| 3          | Next element is a price |
| 100        | Price                   |

Below are examples of getdata reply files in which bulk information is requested using the OUTPUTFORMAT=bulklist (call schedule and index members):

```
START-OF-FILE  
RUNDATE=20050509  
PROGRAMFLAG=oneshot  
FIRMNAME=firmabc  
REPLYFILENAME=bulktest.out  
OUTPUTFORMAT=bulklist  
SECMaster=yes  
PROGRAMNAME=getdata  
DATEFORMAT=yyyymmdd
```

```
START-OF-FIELDS  
CALL_SCHEDULE  
END-OF-FIELDS
```

```
TIMESTARTED=Mon May 9 12:24:20 EDT 2005  
START-OF-DATA  
US00206QAS30 Corp |20050515|104.0600  
US00206QAS30 Corp |20060515|103.6600  
US00206QAS30 Corp |20070515|103.2500  
US00206QAS30 Corp |20080515|102.8400  
US00206QAS30 Corp |20090515|102.4400  
US00206QAS30 Corp |20100515|102.0300  
US00206QAS30 Corp |20110515|101.6300  
US00206QAS30 Corp |20120515|101.2200  
US00206QAS30 Corp |20130515|100.8100  
US00206QAS30 Corp |20140515|100.4100  
US00206QAS30 Corp |20150515|100.0000  
END-OF-DATA  
TIMEFINISHED=Mon May 9 12:24:22 EDT 2005  
END-OF-FILE
```

START-OF-FILE  
RUNDATE=20050509  
PROGRAMFLAG=oneshot  
FIRMNAME=firmabc  
REPLYFILENAME=members.out  
OUTPUTFORMAT=bulklist  
SECMaster=yes  
PROGRAMNAME=getdata

START-OF-FIELDS  
INDX\_MEMBERS  
END-OF-FIELDS

TIMESTARTED=Mon May 9 12:30:35 EDT 2005

START-OF-DATA

|            |         |
|------------|---------|
| INDU Index | AA UN   |
| INDU Index | AIG UN  |
| INDU Index | AXP UN  |
| INDU Index | BA UN   |
| INDU Index | C UN    |
| INDU Index | CAT UN  |
| INDU Index | DD UN   |
| INDU Index | DIS UN  |
| INDU Index | GE UN   |
| INDU Index | GM UN   |
| INDU Index | HD UN   |
| INDU Index | HON UN  |
| INDU Index | HPQ UN  |
| INDU Index | IBM UN  |
| INDU Index | INTC UQ |
| INDU Index | JNJ UN  |
| INDU Index | JPM UN  |
| INDU Index | KO UN   |
| INDU Index | MCD UN  |
| INDU Index | MMM UN  |
| INDU Index | MO UN   |
| INDU Index | MRK UN  |
| INDU Index | MSFT UQ |
| INDU Index | PFE UN  |
| INDU Index | PG UN   |
| INDU Index | SBC UN  |
| INDU Index | UTX UN  |
| INDU Index | VZ UN   |
| INDU Index | WMT UN  |
| INDU Index | XOM UN  |

END-OF-DATA

TIMEFINISHED=Mon May 9 12:30:36 EDT 2005

END-OF-FILE

Clients should be aware that even when specifying OUTPUTFORMAT=bulklist, requesting bulk and non-bulk fields in the same request file will cause bulk fields to return in the standard, semicolon delimited format.

## Special Characters

There is a group of special characters used on the BLOOMBERG PROFESSIONAL™ service to represent fractions that are available to customers by using the SPECIALCHAR option in the header file. These characters can occur in price fields. The characters start with 0x80 and go up sequentially. The list of characters and their translations are:

0x80 - 1/64

0x81 - 1/32

0x82 - 3/64

0x83 - 1/16

...

0xbd - 31/32

0xbe - 63/64

The customer is in no way required to use these characters. These characters (while not printable) are provided so the customer can translate them into their own fraction keys if desired.

## Single-Point History

Single-point history fields allow clients to retrieve data on a given historical date. The following single-point history fields may be used with the getdata program (note that the delimiter here is a colon):

|                      |                   |
|----------------------|-------------------|
| BHIS_CLOSE_ON_PX:n:P | Closing Bid Price |
| BHIS_CLOSE_ON_PX:n:Y | Closing Bid Yield |
| MHIS_CLOSE_ON_PX:n:P | Closing Price     |
| MHIS_CLOSE_ON_PX:n:Y | Closing Yield     |
| AHIS_CLOSE_ON_PX:n:P | Closing Ask Price |
| AHIS_CLOSE_ON_PX:n:Y | Closing Ask Yield |
| HIS_HIGH_ON_PX:n:P   | High Price        |
| HIS_HIGH_ON_PX:n:Y   | High Yield        |
| HIS_LOW_ON_PX:n:P    | Low Price         |
| HIS_LOW_ON_PX:n:Y    | Low Yield         |
| HIS_VOL_ON_PX:n:P    | Volume            |
| N_DAYS_AGO_PX:n      | Date "n" days ago |

Except for N\_DAYS\_AGO\_PX, "n" in the above formats can be either number of days (greater than 0) or a date (excluding today). If "n" is specified as a number of days, it means the number of business days from today excluding weekends. Note the "P" or "Y" at the end of the field indicates a price or yield, respectively. All of these fields fall under the End-of-Day pricing category, not the History category.

For example:

|                             |                       |
|-----------------------------|-----------------------|
| BHIS_CLOSE_ON_PX:5:P        | Bid Price, 5 days ago |
| AHIS_CLOSE_ON_PX:19971217:P | Ask Price on 12/17/97 |

The HIST\_CRNCY header option can be used with these fields to override currency. Please see File Header Section for details.

## **Bond Quote Composite**

The Data License Bond Quote Composite offers a composite view of bids and asks for all contributed pricing sources to which a client is entitled. This program was developed as an efficient service for snapshot valuations of contributed pricing levels for a custom group of bonds.

Bond Quote Composite data is returned within the bulk field **BOND\_QUOTE\_COMP**. Request file headers must contain

**SECMaster=yes**  
**QUOTEComposite=yes**

The field will return, in order, Pricing Source Abbreviation, Pricing Source Name, Time Stamp, Date Stamp, Bid Price, Ask Price, Bid Yield, Ask Yield, Bid Size, Ask Size and Executable Indicator (Y or N).

Data requests are processed upon receipt; the resulting output files are delivered on hourly intervals, provided the request file is uploaded 15 minutes prior. For example, in order for a file to be delivered at 2pm, the request file will need to be posted by 1:45pm. If the file takes longer than 15 minutes to process, it will be delivered once processing is completed.

## **Sample Bond Quote Composite Request File**

```
START-OF-FILE
PROGRAMNAME=getdata
PROGRAMFLAG=one-shot
FIRMNAME=firmabc
QUOTEComposite=YES
SECMaster=YES
OUTPUTFORMAT=bulklist
```

```
START-OF-FIELDS
BOND_QUOTE_COMP
END-OF-FIELDS
```

```
START-OF-DATA
459200AT Corp
END-OF-DATA
END-OF-FILE
```

## Sample Bond Quote Composite Output File

START-OF-FILE  
 PROGRAMNAME=getdata  
 PROGRAMFLAG=one-shot  
 FIRMNAME=firmabc  
 QUOTECOMPOSITE=YES  
 SECMASTER=YES  
 OUTPUTFORMAT=bulklist

START-OF-FIELDS  
 BOND\_QUOTE\_COMP  
 END-OF-FIELDS

START-OF-DATA  
 459200AT Corp|EXCH|EXCHANGE TRADED|09:29:30|4/25/07|99.000|0.000|5.978|0.000|50|0|N  
 459200AT Corp|FTNF|FTN FINANCIAL|10:24:12|4/25/07|0.000|100.765|0.000|4.910|0|88000|N  
 459200AT Corp|DAIN|DAIN RAUSCHER|01:51:40|4/25/07|103.300|0.000|3.416|0.000|30000|0|N  
 459200AT Corp|RBCY|RBC NEW YORK|10:21:37|4/25/07|100.430|100.680|5.110|4.960|500000|19000|N  
 459200AT Corp|MERY|MERRILL LYNCH CORPS|10:22:41|4/25/07|100.810|100.680|4.883|4.960|0|0|N  
 459200AT Corp|NMS|NOMURA SECS|10:24:18|4/25/07|100.365|100.455|5.149|5.095|0|0|N  
 459200AT Corp|MLIX|MERRILL LYNCH FI IDX|00:00:00|4/24/07|100.571|0.000|5.026|0.000|0|0|N  
 459200AT Corp|NYSE|NEW YORK STOCK EXCH|00:00:00|4/25/07|99.000|0.000|5.978|0.000|50|0|N  
 459200AT Corp|TRAC|NASD - TRACE|09:43:09|4/25/07|100.752|100.752|4.917|4.917|0|0|N  
 459200AT Corp|DREU|DRESDNER BANK|10:24:09|4/25/07|100.575|100.658|5.023|4.973|0|0|N  
 459200AT Corp|BDIR|Jefferies-BD|10:12:31|4/25/07|100.348|0.000|5.160|0.000|250000|0|N  
 459200AT Corp|GMSO|GMS GROUP INC|10:23:10|4/25/07|100.170|101.003|5.267|4.767|250000|250000|N  
 459200AT Corp|RBCZ|RBC TORONTO.|10:21:37|4/25/07|100.430|100.680|5.110|4.960|500000|19000|N  
 459200AT Corp|TRST|NASD TRACE (<=1MM)|09:43:09|4/25/07|100.752|100.752|4.917|4.917|0|0|N  
 459200AT Corp|TRL1|NASD TRACE (>1MM)|00:00:00|4/20/07|100.558|100.558|5.033|5.033|0|0|N  
 459200AT Corp|TRMT|NASD TRACE (>=250M)|00:00:00|4/23/07|100.767|100.767|4.908|4.908|0|0|N  
 459200AT Corp|MSRX|MORGAN STANLEY GWMG|10:23:16|4/25/07|0.000|100.608|0.000|5.003|0|10000|N  
 459200AT Corp|SNY1|SIAC - NYSE 1|00:00:00|4/25/07|99.000|0.000|5.978|0.000|50|0|N  
 459200AT Corp|BADR|BoA Securities|10:21:06|4/25/07|100.575|0.000|5.023|0.000|500000|0|N  
 459200AT Corp|FTID|FT INTERACTIVE DATA|00:00:00|4/24/07|100.571|100.571|5.025|5.025|0|0|N  
 459200AT Corp|BGN|BLOOMBERG GENERIC|18:05:43|4/24/07|100.539|100.789|5.045|4.895|0|0|N  
 459200AT Corp|BFV|BLOOMBERG FAIR VALUE|10:24:22|4/25/07|100.463|100.463|5.090|5.090|0|0|N  
 END-OF-DATA  
 END-OF-FILE

## Sample Request File for the getdata Program

```
START-OF-FILE  
FIRMNAME=firmabc  
REPLYFILENAME=getdatatest.out  
SECMaster=yes  
PROGRAMNAME=getdata
```

```
START-OF-FIELDS  
NAME  
TICKER  
CPN  
MATURITY  
END-OF-FIELDS
```

```
START-OF-DATA
```

```
#Requesting the T14 Govt bond in 4 different ways  
912810CY2 |CUSIP  
US912810CY20 | ISIN  
000863149 | VALOREN  
T14 11/15/11 Govt
```

```
#Coupon and Maturity do not apply to equities, so blanks are  
returned  
IBM US Equity
```

```
#FirmABC is not a valid security, return code 10 and NO data is  
returned  
FirmABC Equity  
END-OF-DATA  
END-OF-FILE
```

## Sample Reply File for the getdata Program

```
START-OF-FILE
RUNDATE=20050509
PROGRAMFLAG=oneshot
FIRMNAME=firmabc
REPLYFILENAME=getdatatest.out
SECMaster=yes
PROGRAMNAME=getdata

START-OF-FIELDS
NAME
TICKER
CPN
MATURITY
END-OF-FIELDS

TIMESTARTED=Mon May  9 12:34:47 EDT 2005
START-OF-DATA
#Requesting the T14 Govt bond in 4 different ways
912810CY2|0|4|US TREASURY N/B|T|14.000000|11/15/11|
US912810CY20|0|4|US TREASURY N/B|T|14.000000|11/15/11|
000863149|0|4|US TREASURY N/B|T|14.000000|11/15/11|
T14 11/15/11 Govt|0|4|US TREASURY N/B|T|14.000000|11/15/11|
#Coupon and Maturity do not apply to equities, so blanks are
returned
IBM US Equity|0|4|INTL BUSINESS MACHINES CORP|IBM| | |
#FirmABC is not a valid security, return code 10 and NO data is
returned
FirmABC Equity|10|4| | | | |
END-OF-DATA
TIMEFINISHED=Mon May  9 12:34:49 EDT 2005
END-OF-FILE
```

## Gethistory

This program retrieves various historical data fields for the specified list of securities within the given date range. The program can retrieve up to approximately 6000 points (6000 prices if a price field is requested) of data (approximately 20 years of daily data, trading 5 days a week). Fields are listed between the START-OF-FIELDS and END-OF-FIELDS keywords in the request file. For a list of available data fields, see the History Fields table on the following page.

For equities, by default, the gethistory program will adjust volume and price history for splits. Historical prices are not adjusted for cash dividends. Clients have the option of changing these parameters by [linking their request files](#) to a Bloomberg terminal in their firm.

Please see the [DATERANGE](#) section for instructions on establishing a date range. The format of a gethistory output file is determined by the [HIST\\_FORMAT](#) option; example of both format are shown below.

The gethistory program can be used in conjunction with wildcards. It is thus possible to ask for bid price, ask price and last trade for all stocks that trade on the London stock exchange from 1990 January 1 to today.

**Output files for the gethistory program will be COMPRESSED. If COMPRESS=yes is not in the request file, it will be automatically added. Files may be decompressed using the UNIX tool gunzip or WinZip for PC applications.**

### Return Codes:

- 0 Good return. No errors occurred.
- 14 Field is not recognized or supported by the gethistory program.
- 13 Field does not apply to security.
- 12 Field is not available.
- 10 Start date > End date.
- 10 Bloomberg cannot find the security as specified.
- 990 System Error ([Contact Technical Support](#))
- 994 User does not have permission (contractual) to download history for this security.
- 995 Maximum number of fields exceeded.
- 996 Buffer Overflow (some data for this security is missing).
- 998 Security identifier type (e.g. CUSIP) is not recognized.
- 999 System Error ([Contact Technical Support](#))



**Applicable gethistory file header options**

|             |               |            |
|-------------|---------------|------------|
| COMPRESS    | HIST_PERIOD   | RUNDATE    |
| DATEFORMAT  | HIST_OPTION   | SECID      |
| DATERANGE   | LAUNCH        | SN         |
| FILETYPE    | LOGIN         | SYSTEM     |
| FIRMNAME    | PORTSECDES    | TIME       |
| HEADER      | PROGRAMFLAG   | USERNUMBER |
| FILETYPE    | PROGRAMNAME   | VERSION    |
| HEADER      | PRP           | WS         |
| HIST_CRNCY  | REPLYFILENAME | YELLOWKEY  |
| HIST_FORMAT | REPORT        |            |

**History Fields**

The following table is a list of fields currently available for the “gethistory” program. For a more detailed explanation of the fields, please see the Data Dictionary.

| Field Mnemonic                  | Field Description                         |
|---------------------------------|-------------------------------------------|
| ASSET_SWAP_SPD_BID              | Bid Asset Swap                            |
| ASSET_SWAP_SPD_MID              | Mid Asset Swap                            |
| BN_SURVEY_AVERAGE               | BN Survey Average Price                   |
| BN_SURVEY_HIGH                  | BN Survey High Price                      |
| BN_SURVEY_LOW                   | BN Survey Low Price                       |
| BN_SURVEY_MEDIAN                | BN Survey Median Price                    |
| CHG_NET_1D                      | Change 1 Day Net                          |
| CHG_NET_2D                      | Change 2 Day Net                          |
| CHG_NET_5D                      | Change 5 Day net                          |
| CNVX_OAS_BID                    | Bid OAS                                   |
| CUR_MKT_CAP                     | Current Market Cap                        |
| DUR_ADJ_OAS_BID                 | Bid OAS Effective Duration                |
| EQY_DVD_YLD_12M                 | Gross 12 Month Dividend Yield             |
| EQY_SH_OUT                      | Shares Outstanding                        |
| EQY_TURNOVER                    | Equity Turnover                           |
| EQY_WEIGHTED_AVG_PX             | Volume Weighted Average Price             |
| FUND_CLASS_ASSETS               | Fund Class Assets                         |
| FUND_NET_ASSET_VAL              | Fund Net Asset Value                      |
| FUND_TOTAL_ASSETS               | Fund Total Assets                         |
| FUT_NOTL_CNV_YLD                | Conventional Notational Yield             |
| GERMAN_TOT_VOL_INCL_BANK_TRADES | German Total Volume Including Bank Trades |
| HIST_CALL_IMP_VOL               | Historical Call Implied Volatility        |
| HIST_PUT_IMP_VOL                | Historical Put Implied Volatility         |

|                           |                                        |
|---------------------------|----------------------------------------|
| LAST_TRADE_ONLY           | Last Actual Trade                      |
| MF_BLK_1D                 | Block Money Flow                       |
| MF_NONBLK_1D              | Non-Block Money Flow                   |
| MMKT_7D_YIELD             | Money Market Seven Day Yield           |
| MMKT_AVG_DAYS_TO_MATURITY | Money Market Average Days to Maturity  |
| MOV_AVG_200D              | 200 Day Moving Average                 |
| MOV_AVG_30D               | 30 Day Moving Average                  |
| MOV_AVG_60D               | 60 Day Moving Average                  |
| OAS_SPREAD_BID            | Bid OAS Spread                         |
| OAS_VOL_BID               | Bid OAS Volatility                     |
| OPEN_INT                  | Open Interest                          |
| PE_RATIO                  | Price/Earnings Ratio                   |
| PX_ASK                    | Ask Price                              |
| PX_AT_TRADE_VOLUME        | AT Trade Volume for London Sets Stocks |
| PX_BID                    | Bid Price                              |
| PX_CANCELLATION           | Cancellation Price                     |
| PX_FIXING                 | Fixing Price                           |
| PX_HIGH                   | High Price                             |
| PX_LAST                   | Last Price                             |
| PX_LONDON_MANUAL_VOLUME   | London Manual Trade Volume             |
| PX_LOW                    | Low Price                              |
| PX_LOW_LEGACY             | Low Legacy Price                       |
| PX_MID                    | Mid Price                              |
| PX_NASDAQ_CLOSE           | NASDAQ Closing Price                   |
| PX_OFF_EXCH_VOLUME        | Off Exchange Volume                    |
| PX_OPEN                   | Open Price                             |
| PX_TO_BOOK_RATIO          | Price to Book Ratio                    |
| PX_TO_CASH_FLOW           | Price to Cash Flow                     |
| PX_TO_SALES_RATIO         | Price to Sales Ratio                   |
| PX_VOLUME                 | Volume                                 |
| RSI_14D                   | 14 Day RSI                             |
| RSI_30D                   | 30 Day RSI                             |
| RSI_3D                    | 3 Day RSI                              |
| RSI_9D                    | 9 Day RSI                              |
| TED_IMP_PX_SPD            | TED Implied Price Spread               |
| VOLUME_TOTAL_CALL         | Total Call Volume                      |
| VOLUME_TOTAL_PUT          | Total Put Volume                       |
| YLD_ANNUAL_ASK            | Ask Annual Yield                       |
| YLD_ANNUAL_BID            | Bid Annual Yield                       |
| YLD_ANNUAL_MID            | Mid Annual Yield                       |
| YLD_BLENDED_ASK           | Ask Blended Yield                      |
| YLD_BLENDED_BID           | Bid Blended Yield                      |
| YLD_BLENDED_MID           | Mid Blended Yield                      |
| YLD_CMPD_JPN_ASK          | Ask Japan Compounded Yield             |
| YLD_CMPD_JPN_BID          | Bid Japan Compounded Yield             |
| YLD_CMPD_JPN_MID          | Mid Japan Compounded Yield             |
| YLD_CNV_ASK               | Ask Yield to Worst                     |
| YLD_CNV_BID               | Bid Yield to Worst                     |
| YLD_CNV_MID               | Mid Yield to Worst                     |

|                     |                             |
|---------------------|-----------------------------|
| YLD_CUR_ASK         | Ask Current Yield           |
| YLD_CUR_BID         | Bid Current Yield           |
| YLD_CUR_MID         | Mid Current Yield           |
| YLD_SEMI_ANNUAL_ASK | Ask Semi-Annual Yield       |
| YLD_SEMI_ANNUAL_BID | Bid Semi-Annual Yield       |
| YLD_SEMI_ANNUAL_MID | Mid Semi-Annual Yield       |
| YLD_SOV_SPREAD_ASK  | Ask Sovereign Spread        |
| YLD_SOV_SPREAD_BID  | Bid Sovereign Spread        |
| YLD_SOV_SPREAD_MID  | Mid Sovereign Spread        |
| YLD_SPL_JPN_ASK     | Ask Simple Yield (Japanese) |
| YLD_SPL_JPN_BID     | Bid Simple Yield (Japanese) |
| YLD_SPL_JPN_MID     | Mid Simple Yield (Japanese) |
| YLD_STR_ASK         | Ask Stripped Yield          |
| YLD_STR_BID         | Bid Stripped Yield          |
| YLD_STR_MID         | Mid Stripped Yield          |
| YLD_YTC_ASK         | Ask Yield to Next Call      |
| YLD_YTC_BID         | Bid Yield to Next Call      |
| YLD_YTC_MID         | Mid Yield to Next Call      |
| YLD_YTM_ASK         | Ask Yield to Maturity       |
| YLD_YTM_BID         | Bid Yield to Maturity       |
| YLD_YTM_MID         | Mid Yield to Maturity       |

### Vertical Output Format:

```

START SECURITY | <Identifier> | <Field mnemonic> |
<Identifier>| <Date-1> | <Value-1> |
<Identifier>| <Date-2> | <Value-2> |
...
<Identifier> |<Date-N> | <Value-N>|
END SECURITY | <Identifier> | <Field mnemonic> | <Rcode> |

```

|                       |                                                   |
|-----------------------|---------------------------------------------------|
| <Identifier>          | Security identifier                               |
| <Field mnemonic >     | Requested field                                   |
| <Rcode >              | Return code                                       |
| <Date-1>...<Date-N>   | Date of data element #1...Date of data element #N |
| <Value-1>...<Value-N> | Data element #1...Data element #N                 |

## **Sample Request File for the gethistory Program with default Vertical format**

```
START-OF-FILE
FIRMNAME=firmabc
# Compression is done automatically for ALL gethistory requests
COMPRESS=yes
REPLYFILENAME=testhistorypc.out
DATERANGE=20050502|20050506
PROGRAMNAME=gethistory

START-OF-FIELDS
PX_ASK
PX_BID
END-OF-FIELDS

START-OF-DATA
US912810CY20|ISIN
INTC US Equity
END-OF-DATA
END-OF-FILE
```

## Sample Reply File for the gethistory Program with default Vertical format

```
START-OF-FILE
RUNDATE=20050519
PROGRAMFLAG=oneshot
FIRMNAME=firmabc
# Compression is done automatically for ALL gethistory requests
COMPRESS=yes
REPLYFILENAME=testhistorypc.out
DATERANGE=20050502|20050506
PROGRAMNAME=gethistory

START-OF-FIELDS
PX_ASK
PX_BID
END-OF-FIELDS

TIMESTARTED=Thu May 19 10:34:29 EDT 2005
START-OF-DATA
START SECURITY|US912810CY20 |    PX_BID |
US912810CY20|05/02/2005|115.578125|
US912810CY20|05/03/2005|115.515625|
US912810CY20|05/04/2005|115.546875|
US912810CY20|05/05/2005|115.578125|
US912810CY20|05/06/2005|115.265625|
END SECURITY|US912810CY20 |    PX_BID | 0 |
START SECURITY|US912810CY20 |    PX_ASK |
US912810CY20|05/02/2005|115.640625|
US912810CY20|05/03/2005|115.578125|
US912810CY20|05/04/2005|115.609375|
US912810CY20|05/05/2005|115.640625|
US912810CY20|05/06/2005|115.328125|
END SECURITY|US912810CY20 |    PX_ASK | 0 |
START SECURITY|INTC US Equity |    PX_BID |
INTC US Equity|05/02/2005|23.55|
INTC US Equity|05/03/2005|23.83|
INTC US Equity|05/04/2005|24.14|
INTC US Equity|05/05/2005|24.25|
INTC US Equity|05/06/2005|24.47|
END SECURITY|INTC US Equity |    PX_BID | 0 |
START SECURITY|INTC US Equity |    PX_ASK |
INTC US Equity|05/02/2005|23.56|
INTC US Equity|05/03/2005|23.81|
INTC US Equity|05/04/2005|24.11|
INTC US Equity|05/05/2005|24.26|
INTC US Equity|05/06/2005|24.48|
END SECURITY|INTC US Equity |    PX_ASK | 0 |
END-OF-DATA
TIMEFINISHED=Thu May 19 10:34:30 EDT 2005
END-OF-FILE
```

## Horizontal Output Format:

START-OF-DATA

```
| <Identifier> | <Rcode> | <NumFields> | <Date-1> | <Value-1> | ... <Value-N> |  
| <Identifier> | <Rcode> | <NumFields> | <Date-2> | <Value-1> | ... <Value-N> |
```

...

```
| <Identifier> | <Rcode> | <NumFields> | <Date-N> | <Value-1> | ... <Value-N> |  
END-OF-DATA
```

|                       |                                                   |
|-----------------------|---------------------------------------------------|
| <Identifier>          | Security identifier                               |
| <Rcode >              | Return code                                       |
| <NumFields>           | Number of requested fields                        |
| <Date-1>...<Date-N>   | Date of data element #1...Date of data element #N |
| <Value-1>...<Value-N> | Data element #1...Data element #N                 |

```
| <Identifier> | <Rcode> | <NumFields> | <Date-1> | <Value-1> | ... <Value-N> |
```

## Sample Request File for the gethistory Program in horizontal format

START-OF-FILE

FIRMNAME=firmabc

# Compression is done automatically for ALL gethistory requests

COMPRESS=yes

REPLYFILENAME=testhistorypc.out

DATERANGE=20050502|20050506

# HIST\_FORMAT has been set to horizontal

HIST\_FORMAT=horizontal

PROGRAMNAME=gethistory

START-OF-FIELDS

PX\_ASK

PX\_BID

END-OF-FIELDS

START-OF-DATA

US912810CY20|ISIN

INTC US Equity

END-OF-DATA

END-OF-FILE

## Sample Reply File for the gethistory Program in horizontal format

```
START-OF-FILE
RUNDATE=20050519
PROGRAMFLAG=oneshot
FIRMNAME=firmabc
# Compression is done automatically for ALL gethistory requests
COMPRESS=yes
REPLYFILENAME=testhistorypc.out
DATERANGE=20050502|20050506
# HIST_FORMAT has been set to horizontal
HIST_FORMAT=horizontal
PROGRAMNAME=gethistory

START-OF-FIELDS
PX_ASK
PX_BID
END-OF-FIELDS

TIMESTARTED=Thu May 19 10:36:01 EDT 2005
START-OF-DATA
US912810CY20|0|2|05/02/2005|115.578125|115.640625|
US912810CY20|0|2|05/03/2005|115.515625|115.578125|
US912810CY20|0|2|05/04/2005|115.546875|115.609375|
US912810CY20|0|2|05/05/2005|115.578125|115.640625|
US912810CY20|0|2|05/06/2005|115.265625|115.328125|
INTC US Equity|0|2|05/02/2005|23.550000|23.560000|
INTC US Equity|0|2|05/03/2005|23.830000|23.810000|
INTC US Equity|0|2|05/04/2005|24.140000|24.110000|
INTC US Equity|0|2|05/05/2005|24.250000|24.260000|
INTC US Equity|0|2|05/06/2005|24.470000|24.480000|
END-OF-DATA
TIMEFINISHED=Thu May 19 10:36:02 EDT 2005
END-OF-FILE
```

## Getticks

This program returns every last sale (price level at which trades were executed) time stamped with date, hour, minute and second. Bloomberg keeps a maximum of 50 trading days of tick data, provided the security has been priced for that amount of time. The getticks program ignores any fields specified between the keywords START-OF-FIELDS and END-OF-FIELDS. Note that the first data point for the getticks program is the most recent tick, followed by older ticks.

Please see the **DATERANGE** section for instructions on establishing a date range. Many data points can potentially be returned for each date. The getticks program can be used in conjunction with wildcards.

Output files for the getticks program will be COMPRESSED. If COMPRESS=yes is not in the request file, it will be automatically added. Files may be decompressed using the UNIX tool gunzip or WinZip on a PC.

Format: START SECURITY | <Identifier>  
 <Identifier> | <Month-1/Day-1> | <Time-1> | <Price-1> | <Volume-1> | <Tick code-1> |  
 <Condition code-1> |  
 <Identifier> | <Month-2/Day-2> | <Time-2> | <Price-2> | <Volume-2> | <Tick code-2> |  
 <Condition code-2> |  
 ...  
 <Identifier> | <Month-N/Day-N> | <Time-N> | <Price-N> | <Volume-N> | <Tick code-N> |  
 <Condition code-N> |  
 END SECURITY | <Identifier> | <Field mnemonic> | <Rcode> |

| Bloomberg Field                         | Description                                           |
|-----------------------------------------|-------------------------------------------------------|
| <Identifier>                            | Security identifier                                   |
| <Month-1/Day-1>...<Month-N/Day-N>       | Month/Day of tick #1...Month/Day of tick #N           |
| <Time-1>...<Time-N>                     | Time of tick #1...Time of tick #N                     |
| <Price-1>...<Price-N>                   | Price of tick #1...Price of tick #N                   |
| <Volume-1>...<Volume-N>                 | Volume of tick #1...Volume of tick #N                 |
| <Tick Code-1>...<Tick Code-N>           | Exchange code of tick #1 ... Exchange code of tick #N |
| <Condition code-1>...<Condition code-N> | Condition code of tick #1...Condition code of tick #N |

*Note: The getticks program returns multiple condition codes. Commas will separate the codes within the condition code field.*

### Return Codes:

|     |                                                  |
|-----|--------------------------------------------------|
| 0   | Good return. No errors occurred.                 |
| 10  | Bloomberg cannot find the security as specified. |
| 600 | Error retrieving tick data.                      |
| 601 | Internal database error.                         |
| 602 | Security not found in tick database.             |
| 603 | Error retrieving tick data.                      |



604 Invalid dates specified.  
 605 Permission denied.  
 990 System Error (Contact **Technical Support**)  
 994 Permission denied.  
 995 Maximum number of fields exceeded.  
 996 Buffer Overflow (some data for this security is missing).  
 998 Security identifier type (e.g., CUSIP) is not recognized.  
 999 System Error (Contact **Technical Support**)

### Applicable getticks file header options

|                    |                       |                       |
|--------------------|-----------------------|-----------------------|
| <b>COMPRESS</b>    | <b>PROGRAMNAME</b>    | <b>TICKEXCHLENGTH</b> |
| <b>DATERANGE</b>   | <b>PRP</b>            | <b>TICKLOCALTZ</b>    |
| <b>FILETYPE</b>    | <b>REPLYFILENAME</b>  | <b>TIME</b>           |
| <b>FIRMNAME</b>    | <b>REPORT</b>         | <b>USERNUMBER</b>     |
| <b>HEADER</b>      | <b>RUNDATE</b>        | <b>VERSION</b>        |
| <b>LAUNCH</b>      | <b>SECID</b>          | <b>WS</b>             |
| <b>LOGIN</b>       | <b>SN</b>             | <b>YELLOWKEY</b>      |
| <b>PORTSECDES</b>  | <b>SYSTEM</b>         |                       |
| <b>PROGRAMFLAG</b> | <b>TICKADJUSTDATE</b> |                       |

### Sample Request File for the getticks Program

```
START-OF-FILE
FIRMNAME=firmabc
COMPRESS=yes
REPLYFILENAME=TestGetticks.out
DATERANGE=20050502|20050503
PROGRAMNAME=getticks
```

```
START-OF-DATA
# Microsoft Ticks - will use DATERANGE from file header
MSFT US Equity
# Invalid dates: start date after end date
INTC US Equity||20050506|20050505
# Intel Ticks with different date range than specified in header
INTC US Equity||20050504|20050505
END-OF-DATA
END-OF-FILE
```

## Sample Reply File for the getticks Program

```
START-OF-FILE
RUNDATE=20050509
PROGRAMFLAG=oneshot
FIRMNAME=firmabc
COMPRESS=yes
REPLYFILENAME=TestGetticks.out
DATERANGE=20050502|20050503
PROGRAMNAME=getticks

TIMESTARTED=Mon May 9 12:44:00 EDT 2005
START-OF-DATA
# Microsoft Ticks - will use DATERANGE from file header
START SECURITY|MSFT US Equity|
MSFT US Equity|05/03|18:40:01|25.36||Q|CC|
MSFT US Equity|05/03|18:40:01|25.34|||OC|
MSFT US Equity|05/03|18:40:01|25.2301||Q|OC|
MSFT US Equity|05/03|18:40:01|25.36||Q|NC|
MSFT US Equity|05/03|18:40:01|25.35||Q|OC|
MSFT US Equity|05/03|18:40:01|25.35||D|OC|
...
MSFT US Equity|05/02|07:30:25|25.35|700|Q|FT|
MSFT US Equity|05/02|07:30:25|25.35|300|Q|FT|
MSFT US Equity|05/02|07:30:25|25.35|700|Q|FT|
MSFT US Equity|05/02|07:30:25|25.35|300|Q|FT|
MSFT US Equity|05/02|07:15:58|25.3|2000|Q|FT|
MSFT US Equity|05/02|07:15:01|25.3|1000|Q|FT|
END SECURITY|MSFT US Equity | 0 |
# Invalid dates: start date after end date
START SECURITY|INTC US Equity|
END SECURITY|INTC US Equity | 604 |
# Intel Ticks with different date range than specified in header
START SECURITY|INTC US Equity|
INTC US Equity|05/05|18:40:00|24.26||Q|CC|
INTC US Equity|05/05|18:40:00|24.3|||OC|
INTC US Equity|05/05|18:40:00|24.22||Q|OC|
INTC US Equity|05/05|18:40:00|24.26||Q|NC|
INTC US Equity|05/05|18:40:00|24.25||Q|NC|
INTC US Equity|05/05|18:40:00|24.22||D|OC|
...
INTC US Equity|05/04|08:11:21|23.86|454|Q|FT|
INTC US Equity|05/04|08:11:21|23.85|3000|Q|FT|
INTC US Equity|05/04|08:07:17|23.8|2000|Q|FT|
INTC US Equity|05/04|08:05:45|23.78|3000|Q|FT|
INTC US Equity|05/04|08:05:36|23.78|1000|Q|FT|
INTC US Equity|05/04|08:05:35|23.75|1000|Q|FT|
INTC US Equity|05/04|08:02:04|23.73|46500|Q|FD|
END SECURITY|INTC US Equity | 0 |
END-OF-DATA
TIMEFINISHED=Mon May 9 12:56:58 EDT 2005
END-OF-FILE
```

## Getallticks

This program is very similar to the getticks program except that in addition to returning every last sale (price level at which trades were executed), matching ask and bid prices are returned, time stamped with date, hour, minute and second. (For London equities, mid prices are also returned). This program is available for equities, preferreds and currencies. The getallticks output format for other security types may default to that of the getticks program; for example, this is the getallticks output for a security listed under the corporate database:

```
START SECURITY|ED000891 Corp|
ED000891 Corp|05/23|17:16:07|82.66|||OC|
ED000891 Corp|05/23|16:34:49|84.16|10000|||
ED000891 Corp|05/23|16:34:49|82.66|10000|||
ED000891 Corp|05/23|11:19:19|86.883|9000|||
ED000891 Corp|05/23|11:19:09|86.883|18000|||
END SECURITY|ED000891 Corp | 0 |
```

To receive NASDAQ Market Maker codes, clients must link their data license file to a BLOOMBERG PROFESSIONAL™ service with Nasdaq Level II subscription. In addition, certain exchanges mandate that a client must be subscribed to real-time pricing for the given exchange in order to download bid and ask prices; an example is the New York Stock Exchange. Please reference the section on linking to a Bloomberg terminal.

For the format on trades, please see the getticks program. For Mid prices on London equities, the format is the same as for trades except that the <Condition code> may be 'Mid' or 'OC' (official close).

**Output files for the getallticks program will be COMPRESSED. If COMPRESS=yes is not in the request file, it will be automatically added. Files may be decompressed using the UNIX tool gunzip or, on a PC, WinZip.**

### *Bid and Ask ticks*

#### **Format:**

```
START SECURITY | <Identifier>
<Identifier> | <Month-1/Day-1> | <Time-1> |
<TypeB-1> | <PriceB-1> | <Tick CodeB-1> | <VolumeB-1> |
<TypeA-1> | <PriceA-1> | <Tick CodeA-1> | <VolumeA-1> | <Condition code-1> |
<Identifier> | <Month-2/Day-2> | <Time-2> |
<TypeB-2> | <PriceB-2> | <Tick CodeB-2> | <VolumeB-2> |
<TypeA-2> | <PriceA-2> | <Tick CodeA-2> | <VolumeA-2> | <Condition code-2> |
...
<Identifier> | <Month-N/Day-N> | <Time-N> |
<TypeB-N> | <PriceB-N> | <Tick CodeB-N> | <VolumeB-N> |
<TypeA-N> | <PriceA-N> | <Tick CodeA-N> | <VolumeA-N> | <Condition code-N> |
END SECURITY | <Identifier> | <Rcode> |
```

### Trade ticks

```
<Identifier> | <Month-1/Day-1> | <Time-1> | <Price-1> | <Volume-1> | <Tick code-1> |
<Condition code-1>|
<Identifier> | <Month-2/Day-2> | <Time-2> | <Price-2> | <Volume-2> | <Tick code-2> |
<Condition code-2>|
...
<Identifier> | <Month-N/Day-N> | <Time-N> | <Price-N> | <Volume-N> | <Tick code-N> |
<Condition code-N>|
```

| Bloomberg Field                         | Description                                                                                                     |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <Identifier>                            | Security identifier                                                                                             |
| <Month-1/Day-1>...<Month-N/Day-N>       | Month/Day of tick #1...Month/Day of tick #N                                                                     |
| <TimeB-1>...<TimeB-N>                   | Time of bid tick #1...Time of bid tick #N                                                                       |
| <TypeB-1>...<TypeB-N>                   | 'B' (bid price)                                                                                                 |
| <PriceB-1>...<PriceB-N>                 | Price of bid tick #1...Price of bid tick #N                                                                     |
| <Tick CodeB-1>...<Tick CodeB-N>         | Exchange code/market maker code for bid tick #1... Exchange code/market maker code for bid tick #N <sup>1</sup> |
| <VolumeB-1>...<VolumeB-N>               | Volume of bid tick #1...Volume of bid tick #N                                                                   |
| <TypeA-1>...<TypeA-N>                   | 'A' (ask price)                                                                                                 |
| <PriceA-1>...<PriceA-N>                 | Price of ask tick #1...Price of ask tick #N                                                                     |
| <Tick CodeA-1>...<Tick CodeA-N>         | Exchange code/market maker code for ask tick #1... Exchange code/market maker code for ask tick #N <sup>1</sup> |
| <VolumeA-1>...<VolumeA-N>               | Volume of ask tick #1...Volume of ask tick #N                                                                   |
| <Condition code-1>...<Condition code-N> | Condition code of bid and ask ticks #1...Condition code of bid and ask ticks #N                                 |

<sup>1</sup> If market maker codes are provided on the exchange feed, they will be returned; otherwise, exchange codes will return in the output file.

Note: The getallticks program returns multiple condition codes. Commas will separate the codes within the condition code field.

### Return Codes:

|     |                                                           |
|-----|-----------------------------------------------------------|
| 0   | Good return. No errors occurred.                          |
| 10  | Bloomberg cannot find the security as specified.          |
| 600 | Error retrieving tick data.                               |
| 601 | Internal database error.                                  |
| 602 | Security not found in tick database.                      |
| 603 | Error retrieving tick data.                               |
| 604 | Invalid dates specified.                                  |
| 605 | Permission denied.                                        |
| 650 | Invalid security type.                                    |
| 651 | Internal database error.                                  |
| 990 | System Error (Contact <a href="#">Technical Support</a> ) |

### Applicable getallticks file header options

|             |                |                |
|-------------|----------------|----------------|
| COMPRESS    | PROGRAMNAME    | TICKEXCHLENGTH |
| DATERANGE   | PRP            | TICKLOCALTZ    |
| FILETYPE    | REPLYFILENAME  | TIME           |
| FIRMNAME    | REPORT         | USERNUMBER     |
| HEADER      | RUNDATE        | VERSION        |
| LAUNCH      | SECID          | WS             |
| LOGIN       | SN             | YELLOWKEY      |
| PORTSECDES  | SYSTEM         |                |
| PROGRAMFLAG | TICKADJUSTDATE |                |

### Sample Request File for the getallticks Program

```

START-OF-FILE
FIRMNAME=firmabc
COMPRESS=yes
REPLYFILENAME=TestGetallticks.out
DATERANGE=20050505|20050506
PROGRAMNAME=getallticks

```

```

START-OF-DATA
# Yahoo ticks
YH00 US Equity
# Unknown security
FirmABC Equity
# Invalid dates - start date after end date
INTC US Equity||20050502|20050501
END-OF-DATA
END-OF-FILE

```

## Sample Reply File for the getallticks Program

```
START-OF-FILE
RUNDATE=20050519
PROGRAMFLAG=oneshot
FIRMNAME=firmabc
COMPRESS=yes
REPLYFILENAME=TestGetallticks.out
DATERANGE=20050505|20050506
PROGRAMNAME=getallticks

TIMESTARTED=Thu May 19 11:27:07 EDT 2005
START-OF-DATA
# Yahoo ticks
START SECURITY|YH00 US Equity|
YH00 US Equity|05/06|18:40:01|T|34.52|||CC|
YH00 US Equity|05/06|18:40:01|T|34.51|||OC|
YH00 US Equity|05/06|18:40:01|T|34.63|||OC|
. . .
YH00 US Equity|05/06|13:59:51|T|34.51||100||
YH00 US Equity|05/06|13:59:51|T|34.51||100||
YH00 US Equity|05/06|13:59:51|B|34.51||27|A|34.52||178||
YH00 US Equity|05/06|13:59:51|B|34.51||29|A|34.52||178||
YH00 US Equity|05/06|13:59:51|B|34.51||29|A|34.52||175||
. . .
YH00 US Equity|05/05|07:10:22|B|34.71||20|A|35.59||19||
YH00 US Equity|05/05|06:59:44|B|34.71||20|A|35.59||20||
YH00 US Equity|05/05|06:59:32|T|34.71||20||
END SECURITY|YH00 US Equity | 0 |
# Unknown security
START SECURITY|FirmABC Equity|
END SECURITY|FirmABC Equity | 10 |
# Invalid dates - start date after end date
START SECURITY|INTC US Equity|
END SECURITY|INTC US Equity | 604 |
END-OF-DATA
TIMEFINISHED=Thu May 19 11:31:31 EDT 2005
END-OF-FILE
```

## Getactions

For each security in the request file, the getactions program will return corporate actions that apply to that security and its issuer. For example, specifying **IBM US Equity** in a getactions request file will return actions specific to that common stock (e.g., a cash dividend) as well as actions specific to International Business Machines (e.g., an acquisition). The getactions program can be used in conjunction with wildcards.

By default, Bloomberg matches the securities in the request file with corporate actions that were entered into the Bloomberg databases in the 24 hours previous to the start of a request file's processing. The ACTIONS\_DATE option can be used to match the securities against corporate actions that became effective on the date of the request.

Additional options for Corporate Actions:

- **ACTIONS** - used to specify action types or action categories.
- **DATERANGE** - can be used to retrieve actions (based on entry date, effective date, or both) up to a maximum of seven days previous to the request date. A "day" is in a New York time day, midnight to midnight. Additionally, actions that will become effective on a future date can be requested. ACTIONS\_DATE will need to be set to effective for future date requests. *NOTE: The output for a future date request will contain only those actions available at the time of the request.* Additional actions may be entered into the Bloomberg Corporate Action database after the output is provided.

For details about the above options, please see the File Header Section.

Corporate actions are divided into three categories: corporate events, capital change and distributions. The Data License Corporate Actions Reference Guide provides a list of all corporate actions that fall under these categories. The guide also provides a list of data items (fields) returned for each action together with definitions, field mnemonics and field types.

**Please see the file Corporate\_Actions.doc in the Notices folder in the client's home directory for the fields returned for each action type.**

By default, the getactions program will return all applicable corporate actions for the securities listed between START-OF-DATA and END-OF-DATA. However, it is possible to filter on certain corporate actions or categories of corporate actions by using the ACTIONS header variable (see File Header Section). Note: The available categories are CORPORATE\_EVENTS, CAPITAL\_CHANGE and DISTRIBUTIONS.

**Format:**

<Identifier> | <Bloomberg Company ID > | <Bloomberg Security ID> | <Rcode> | <Action ID> | <Mnemonic> | <Flag> | <Company Name> | <SecID-type> | <SecID> | <Currency> | <Market Sector Description> | <Bloomberg Unique ID> | <Ann-date> | <Eff-date> | <Amd-date>| <Nfields> | <fld-mnemo1> | <Value-1> | ... |<fld-mnemoN> | <Value-N> |

The below table includes the standard fields included with every corporate action record, along with a field description, and field information such as field type and standard width:

| Bloomberg Field         | Field Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Field Type<br>(Standard Width) |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <Identifier>            | Security identifier used in the request file                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Character (32)                 |
| <Bloomberg Company ID>  | Number that uniquely identifies a company                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Integer (8)                    |
| <Bloomberg Security ID> | Number that in combinations with Bloomberg Company ID such as Acquisition this field is set to 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Integer (8)                    |
| <Rcode >                | Return code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Integer (4)                    |
| <Action ID>             | A unique action identifier assigned to corporate actions. An update or delete of an action will have the same Action ID as the initial entry of the action. Additionally, acquisitions and divestitures have the same Action ID. These IDs can be seen in the CACT <go> screen of the BLOOMBERG Professional.                                                                                                                                                                                                                                                                                                               | Integer (10)                   |
| <Mnemonic>              | Mnemonic for corporate action                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Character (18)                 |
| <Flag>                  | Status of the action at the moment of the request. Returns N for an action that has been entered and has received no subsequent update, U for an action that has been updated since its entry, and D for a deleted action.<br>NOTE 1: Clients should load actions with the first appearance of an Action ID whether the flag is a N or a U.<br>NOTE 2: If D is returned as a Flag, CP_DELETE_REASON will be provided the only non-standard field for the action. Possible returns are 1 for CANCEL and 2 for MISTAKE.<br>NOTE 3: If an action appears for the first time with a D flag, clients should not load the action. | Character (4)                  |
| <Company Name>          | Name of the company                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Character (80)                 |



|                                 |                                                                                                                                                                                                                                                    |                |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <SecID type>                    | Type of security ID. For Bonds: CUSIP, ISIN, BBID (Bloomberg ID), BB_UNIQUE; For Equities, CUSIP, ISIN, VALOREN, SEDOL, BELGIAN, WPK, SICOVAM, FONDS, BB_UNIQUE. The system will return the first available identifier type within this hierarchy. | Character (16) |
| < SecID>                        | The security ID.                                                                                                                                                                                                                                   | Character (12) |
| < Currency>                     | Currency of security (ISO code)                                                                                                                                                                                                                    | Character (4)  |
| <Market Sector Description>     | The name of the market sector yellow key on the BLOOMBERG™ Professional Service that the security is located under.                                                                                                                                | Character (6)  |
| <Bloomberg Unique ID>           | A unique number assigned by Bloomberg to all securities.                                                                                                                                                                                           | Character (30) |
| < Ann-date>                     | Date when the corporate action is announced.                                                                                                                                                                                                       | Date (10)      |
| < Eff-date>                     | Date when the corporate action becomes effective. The field will return a value of "N.A." for the following actions when they are in "To Be Announced" status (CP_INDICATOR=T):<br>DVD_CASH<br>DVD_STOCK<br>STOCK_SPLT<br>SPIN<br>RIGHTS_OFFER     | Date (10)      |
| <Amd-date>                      | Date when the corporate action was updated. Valid only when <Flag> is "U", otherwise it will be "N.A."                                                                                                                                             | Date (10)      |
| <Nfields>                       | Number of fields.                                                                                                                                                                                                                                  | Integer (4)    |
| < fld-mnemo1 >...< fld-mnemoN > | Mnemonic identifying data elements #1 ... #N                                                                                                                                                                                                       |                |
| <Value-1>...<Value-Nfields>     | Data element #1...#N                                                                                                                                                                                                                               |                |

## Return Codes:

The following return codes are currently defined:

|     |                                                           |
|-----|-----------------------------------------------------------|
| 0   | Good return. No errors occurred.                          |
| 10  | Bloomberg cannot find the security as specified.          |
| 300 | No corporate actions were found for given security.       |
| 990 | System Error (Contact <a href="#">Technical Support</a> ) |
| 994 | Permission denied.                                        |
| 995 | Maximum number of fields exceeded.                        |
| 996 | Buffer Overflow (some data for this security is missing). |
| 998 | Security identifier type (e.g., CUSIP) is not recognized. |
| 999 | System Error (Contact <a href="#">Technical Support</a> ) |

### Applicable getactions file header options

|                     |                      |                   |
|---------------------|----------------------|-------------------|
| <b>ACTIONS</b>      | <b>LAUNCH</b>        | <b>SECID</b>      |
| <b>ACTIONS_DATE</b> | <b>LOGIN</b>         | <b>SN</b>         |
| <b>COMPRESS</b>     | <b>PORTSECDES</b>    | <b>SYSTEM</b>     |
| <b>DATEFORMAT</b>   | <b>PROGRAMFLAG</b>   | <b>TIME</b>       |
| <b>DATERANGE</b>    | <b>PROGRAMNAME</b>   | <b>USERNUMBER</b> |
| <b>FILETYPE</b>     | <b>PRP</b>           | <b>VERSION</b>    |
| <b>FIRMNAME</b>     | <b>REPLYFILENAME</b> | <b>WS</b>         |
| <b>HEADER</b>       | <b>RUNDATE</b>       | <b>YELLOWKEY</b>  |

### Sample Request File for the getactions Program

```

START-OF-FILE
FIRMNAME=firmabc
REPLYFILENAME=TestActions.out
# Looking for Ticker Changes on the specified securities
ACTIONS=CHG_TKR
# Returning actions based on BOTH entry and effective dates
ACTIONS_DATE=both
DATERANGE=20050519|20050526
PROGRAMNAME=getactions

START-OF-DATA
# Three equities and one unknown security
# The two below securities had actions within this date range
2975 HK Equity
KKI LN Equity
# A return code of 300 indicates there is no corporate action for
# this security
IBM Equity
# A return code of 10 indicates that Bloomberg did not recognize
# the security
FIRMABC Equity
END-OF-DATA
END-OF-FILE

```

## Sample Reply File for the getactions Program

Note that each corporate action is returned as one record. There are also no empty lines between corporate actions. The example below has been slightly altered for readability.

```
START-OF-FILE
PROGRAMFLAG=oneshot
FIRMNAME=firmabc
REPLYFILENAME=TestActions.out
# Looking for Ticker Changes on the specified securities
ACTIONS=CHG_TKR
# Returning actions based on BOTH entry and effective dates
ACTIONS_DATE=both
DATERANGE=20050519|20050526
PROGRAMNAME=getactions

TIMESTARTED=Thu May 19 11:43:56 EDT 2005
START-OF-DATA
# Using corporate action software version 2.3.5
# Returning corporate actions effective between 20050519 and
# 20050526, inclusive
# Warning: More actions effective in the future may be entered at
# a later time, including later today.
# Returning corporate actions entered between 00:00 ET on
# 20050519 and # 23:59 ET on 20050526
# Warning: There will not be any actions 'entered in the future'.
# Three equities and one unknown security
# The two below securities had actions within this date range

2975 HK Equity|116526|1000|0|17985085|CHG_TKR|N|Northern
International Holdings Ltd|ISIN|BMG6633T1356|HKD|Equity|
EQ0011652600001000|05/05/2005| 05/20/2005|N.A.|3|CP_OLD_TKR|
2975 HK|CP_NEW_TKR|736 HK|CP_NOTES|N.A.|

KKI LN Equity|125626|1002|0|18121712|CHG_TKR|N|Opus International
Group Plc|ISIN|GB0004935036|GBp|Equity|EQ0012562600001002|
05/19/2005| 05/20/2005|N.A.|3|CP_OLD_TKR|KKI LN|CP_NEW_TKR|
OPU LN|CP_NOTES|N.A.|

# A return code of 300 indicates there is no corporate action for
# this security

IBM Equity|100801|1000|300|

# A return code of 10 indicates that Bloomberg did not recognize
# the security

FIRMABC Equity|0|0|10|
END-OF-DATA
TIMEFINISHED=Thu May 19 11:46:51 EDT 2005
END-OF-FILE
```

## Corporate Actions

Corporate actions are divided into three categories: Corporate Events, Capital Change, and Distributions. A list of data items (fields) returned for each action together with a definition, a field mnemonic, and a field type can be obtained in the Corporate\_Actions.doc file located in the Notices subdirectory of the client's home directory.

**Note: Bloomberg can add new action types as well as new fields to a specific action at any time. Clients will be notified of changes prior to the effective date so that accommodations for code change can be made.**

The chart below lists all available actions for each category:

| Corporate Action Categories   |                                |                |
|-------------------------------|--------------------------------|----------------|
| Corporate Events              | Capital Change                 | Distributions  |
| Name Change                   | Merger                         | Cash Dividends |
| Domicile Change               | Spin-off                       | Stock Dividend |
| State of Incorporation Change | Bankruptcy Filing              | Stock Split    |
| Round Lot Change              | Stock Buyback                  |                |
| Ticker Symbol Change          | Equity Offering                |                |
| ID Number Change              | Debt Redemption Call           |                |
| Equity Delisting              | Debt Offering/Increase         |                |
| Change in Listing             | Debt Offering/New              |                |
| Equity Listing                | Par Value Change               |                |
| Variable Interest Resets      | Debt Redemption Sinker         |                |
| Voting Rights Change          | Acquisition                    |                |
| Currency Quotation Change     | Divestiture                    |                |
| Reconvention                  | Installment Call - Partial Pay |                |
| Redenomination                | Reclassification               |                |
| Shareholder Meeting           | Debt Repurchase                |                |
|                               | Debt Redemption/Put            |                |
|                               | Exchange Offers                |                |
|                               | Variable Principle Redemption  |                |
|                               | Pay in Kind                    |                |
|                               | Rights Offering                |                |
|                               | Conversion Price Refix         |                |
|                               | Exercise Price Refix           |                |
|                               | Funged Issues                  |                |
|                               | Extendible Issues              |                |

The ACTION\_DATE flag can be used with a getactions request to modify whether the reply file will return actions based on effective date, entry date, or both.

## Getcompany

This program returns company/entity level data such as industry classification, country of risk, and country of domicile for a given security/company ID. For the full list of available fields, please see `crisk_fields.csv`, which can be found in the root directory of client FTP servers. **Only those fields listed in this file will be available in this program.** Fields can be specified explicitly or can be requested via the Field Set macros below. The program supports requests by security/ticker or by entity/company ID.

This program requires the inclusion of header option **CREDITRISK=Yes** in the request file. It also requires that **ID\_BB\_COMPANY** be included within the fields list. If **ID\_BB\_COMPANY** is not included, an error report will be generated and no data will be returned.

Format: |<Identifier> |<Rcode> | <Nfields> | <Value-1> | <Value-2> | ....| <Value-Nfields>|

|                              |                                                      |
|------------------------------|------------------------------------------------------|
| <Identifier>                 | Security/Company identifier used in the request file |
| <Rcode>                      | Return Code                                          |
| <Nfields>                    | Number of fields requested and received              |
| <Value-1>....<Value-Nfields> | Data element #1... Data element # Nfields            |

### Return Codes:

|     |                                                                                  |
|-----|----------------------------------------------------------------------------------|
| 0   | Good return. No errors occurred.                                                 |
| 10  | Bloomberg cannot find the security as specified.                                 |
| 11  | Security found, but no associated company                                        |
| 12  | Bloomberg cannot find the security and the identifier syntax/format is incorrect |
| 988 | System Error on security level                                                   |
| 990 | System Error (Contact <b>Technical Support</b> )                                 |
| 994 | Permission denied (Contact DL Sales Rep)                                         |
| 996 | Buffer Overflow (some data for this security is missing).                        |
| 998 | Security identifier type (e.g., CUSIP) is not recognized.                        |
| 999 | System Error (Contact <b>Technical Support</b> )                                 |

**Applicable getcompany file header options**

|                     |                 |                      |                     |                   |
|---------------------|-----------------|----------------------|---------------------|-------------------|
| <b>COLUMNHEADER</b> | <b>FILETYPE</b> | <b>OUTPUTFORMAT</b>  | <b>SECDESLength</b> | <b>TIME</b>       |
| <b>COMPRESS</b>     | <b>FIRMNAME</b> | <b>PORTSECDES</b>    | <b>SECID</b>        | <b>USERNUMBER</b> |
| <b>CREDITRISK</b>   | <b>HEADER</b>   | <b>REPLYFILENAME</b> | <b>SN</b>           | <b>VERSION</b>    |
| <b>DATEFORMAT</b>   | <b>LAUNCH</b>   | <b>REPORT</b>        | <b>SPECIALCHAR</b>  | <b>WS</b>         |
| <b>DELIMITER</b>    | <b>LOGIN</b>    | <b>RUNDATE</b>       | <b>SYSTEM</b>       | <b>YELLOWKEY</b>  |
| <b>FIELDSET</b>     |                 |                      |                     |                   |

**Field Set Macro**

Getcompany has a new option, the FIELDSET macro, which offers the ability to include all fields available in the Credit Risk BackOffice module without explicitly listing each field individually.

Format: FIELDSET=<FieldMacro> | <Date>

**<FieldMacro>**  
**<Date>**

Macro name – corresponds to Credit Risk BackOffice file Optional. This represents the effective date of when the field set was released into production. Setting this field to a specific date will prevent future releases of field sets from impacting existing requests and reply files. Format is YYYYMMDD. Default is the current date if not specified.

The following **Field Macros** are currently available:

|                         |                                             |
|-------------------------|---------------------------------------------|
| BO_CREDIT_RISK_COMPANY  | fields included in credit_risk.out          |
| BO_CREDIT_RISK_FITCH    | fields included in credit_risk_fitch.out    |
| BO_CREDIT_RISK_MOODY    | fields included in credit_risk_moody.out    |
| BO_CREDIT_RISK_RATINGS1 | fields included in credit_risk_ratings1.out |
| BO_CREDIT_RISK_RATINGS2 | fields included in credit_risk_ratings2.out |
| BO_CREDIT_RISK_SP       | fields included in credit_risk_sp.out       |

The list of fields can be found in the *crisklayout.xls* file located in the notices sub-directory on your ftp server.

### **Sample Request File for the getcompany Program (using Security)**

```
START-OF-FILE
FIRMNAME=d112345
REPLYFILENAME=getcompanytest
CREDITRISK=yes
PROGRAMNAME=getcompany
```

```
START-OF-FIELDS
ID_BB_COMPANY
CNTRY_OF_RISK
IS_ULT_PARENT
END-OF-FIELDS
```

```
START-OF-DATA
IBM US Equity
END-OF-DATA
END-OF-FILE
```

### **Sample Reply File for the getcompany Program (using Security)**

```
START-OF-FILE
RUNDATE=20071106
PROGRAMFLAG=oneshot
FIRMNAME=d112345
REPLYFILENAME=getcompanytest
CREDITRISK=yes
PROGRAMNAME=getcompany
```

```
START-OF-FIELDS
ID_BB_COMPANY
CNTRY_OF_RISK
IS_ULT_PARENT
END-OF-FIELDS
```

```
TIMESTARTED=Tue Nov 6 13:03:45 EST 2007
START-OF-DATA
IBM US Equity|0|3|100801|US|Y|
END-OF-DATA
TIMEFINISHED=Tue Nov 6 13:03:47 EST 2007
DATARECORDS=1
END-OF-FILE
```

### **Sample Request File for the getcompany Program (using Company ID)**

```
START-OF-FILE
FIRMNAME=d112345
REPLYFILENAME=getcompany_id_bb
CREDITRISK=yes
SECID=BB_COMPANY
PROGRAMNAME=getcompany
```

```
START-OF-FIELDS
ID_BB_COMPANY
LONG_COMP_NAME
CNTRY_OF_RISK
IS_ULT_PARENT
END-OF-FIELDS
```

```
START-OF-DATA
100801
END-OF-DATA
END-OF-FILE
```

### **Sample Reply File for the getcompany Program (company ID)**

```
START-OF-FILE
RUNDATE=20071106
PROGRAMFLAG=oneshot
FIRMNAME=d112345
REPLYFILENAME=getcompany_id_bb
CREDITRISK=yes
SECID=BB_COMPANY
PROGRAMNAME=getcompany
```

```
START-OF-FIELDS
ID_BB_COMPANY
LONG_COMP_NAME
CNTRY_OF_RISK
IS_ULT_PARENT
END-OF-FIELDS
```

```
TIMESTARTED=Tue Nov 6 13:20:12 EST 2007
START-OF-DATA
100801|0|4|100801|International Business Machines Corp|US|Y|
END-OF-DATA
TIMEFINISHED=Tue Nov 6 13:20:12 EST 2007
DATARECORDS=1
END-OF-FILE
```



**Sample Request File for the getcompany Program using Field Set Macro  
(using Security)**

```
START-OF-FILE
FIRMNAME=dl12345
REPLYFILENAME=getcompany_fld_macro
CREDITRISK=yes
PROGRAMNAME=getcompany

START-OF-FIELDS
FIELDSET=B0_CREDIT_RISK_COMPANY|20071106|
END-OF-FIELDS

START-OF-DATA
IBM US Equity
END-OF-DATA
END-OF-FILE
```

## Sample Reply File for the getcompany Program using Field Set Macro (security)

START-OF-FILE  
RUNDATE=20071106  
PROGRAMFLAG=oneshot  
FIRMNAME=d112345  
REPLYFILENAME=getcompany\_fld\_macro  
CREDITRISK=yes  
SECID=BB\_COMPANY  
PROGRAMNAME=getcompany

START-OF-FIELDS  
##+ FIELDSET=BO\_CREDIT\_RISK\_COMPANY|20071106  
ID\_BB\_COMPANY  
LONG\_COMP\_NAME  
ID\_BB\_PARENT\_CO  
LONG\_PARENT\_COMP\_NAME  
INDUSTRY\_SECTOR  
INDUSTRY\_GROUP  
INDUSTRY\_SUBGROUP  
CNTRY\_OF\_DOMICILE  
CNTRY\_OF\_INCORPORATION  
CNTRY\_OF\_RISK  
STATE\_OF\_DOMICILE  
STATE\_OF\_INCORPORATION  
COMPANY\_ADDRESS  
IS\_ULT\_PARENT  
OBLIG\_INDUSTRY\_SUBGROUP  
ACQUIRED\_BY\_PARENT  
COMPANY\_TO\_PARENT\_RELATIONSHIP  
ISSUER\_NAME\_TYPES  
ID\_BB\_ULTIMATE\_PARENT\_CO  
LONG\_ULT\_PARENT\_COMP\_NAME  
COMPANY\_CORP\_TICKER  
##- FIELDSET=BO\_CREDIT\_RISK\_COMPANY|20071106  
END-OF-FIELDS

TIMESTARTED=Tue Nov 6 13:26:52 EST 2007  
START-OF-DATA  
IBM US Equity|0|21|100801|International Business Machines Corp| |  
|Technology|Computers|Computers|US|US|US|NY|NY|;2;6;1;1;One New Orchard  
Road;1; ;1;Armonk;1;NY;1;10504;1;United States;|Y|Computers|N|  
|Company|100801|International Business Machines Corp|IBM|  
END-OF-DATA  
TIMEFINISHED=Tue Nov 6 13:26:52 EST 2007  
DATARECORDS=1  
END-OF-FILE

## Cross Referencing

### Cross Reference Data - Lookup Values

Lookup fields are used to determine the possible return values for a specific field. For example, a request file containing the field LU\_DAY\_CNT would return the available values for the field DAY\_CNT (Day Count) - 30/360, ACT/360, ACT/365, etc. Please see [Appendix D](#) for more details about accessing Lookup values.

## List of Fields

### Data Dictionary (Fields.csv)

BLOOMBERG Data License provides all clients with the file **fields.csv**. This comma-delimited ASCII file is a comprehensive list of all fields available via Data License. Information provided within this file includes the BLOOMBERG Data License Mnemonic, the Field ID, and a brief description and definition of each field. All Data License fields are assigned a Data License Category.

The table contains 22 columns as described below:

**Field ID:** The Bloomberg 5 character code unique to each mnemonic.

**Field Mnemonic:** The Bloomberg Data License Mnemonic.

**Description:** A short description of each mnemonic.

**Data License Category:** There are currently four Data License categories:

- Security Master
- Derived Data
- End of Day Pricing
- Historical Time Series

**Category:** A general classification of the mnemonic within the Data License Category.

**Definition:** A definition of the mnemonic.

**Comdty, Equity, Muni, Pfd, M-Mkt, Govt, Corp, Index, Curncy, Mtge** - Indicates all applicable market sectors for each mnemonic.

**Standard Width:** The standard width of the return value.

**Standard Decimal Places:** The standard number of decimal places of the return value (if applicable).

**Field Type:** Indicates the data type. Possible field return types are

- Character
- Real
- Integer
- Price
- Bulk
- Data
- Currency
- Boolean
- Date or Time
- Month/Year

**Back Office:** Indicates if the field mnemonic is part of the Back Office Product.

**Production Date:** Indicates the date that the field was placed in production.

**Current Maximum Width:** The maximum width of the return value.

New data fields are continually added to the Data License product. **fields.csv** is updated every Monday; it is suggested that clients download the most recent version of the file from their FTP folder each Monday to ensure that they have a complete list of all available data fields. The file is processed daily except Saturdays.

| Bulk Field Data Element Number | Field Type   | Example                                                                                                                                |
|--------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------|
| 1                              | Character    | 100BP, APPLE COMPUTERS                                                                                                                 |
| 2                              | Numeric      | 55000, 1234.4321                                                                                                                       |
| 3                              | Price        | Format according to SPECIALCHAR flag                                                                                                   |
| 4                              | Security     | IBM US Equity                                                                                                                          |
| 5                              | Date         | Format according to DATEFORMAT flag                                                                                                    |
| 6                              | Time         | hh:mm:ss, 13:25:42                                                                                                                     |
| 7                              | Date or time | Can be either.                                                                                                                         |
| 8                              | Bulk         | A self-defining data type used to encapsulate multi-dimensional (matrix) data. For more information, refer to the Bulk Format section. |
| 9                              | Month/Year   | 12/96                                                                                                                                  |
| 10                             | Boolean      | 'Y' or 'N'                                                                                                                             |
| 11                             | Currency     | USD, United States Dollar                                                                                                              |
| 12                             | Integer      | 1000, 550000, -3                                                                                                                       |
| 13                             | Real         | 3.6, 1234.3340                                                                                                                         |

## **Bloomberg Identifiers**

**ID\_BB\_UNIQUE** – The Bloomberg Unique ID can be used as a key field to link or join securities across files. It is a unique identifier for every security and will never be re-used. It is an automatically generated number for Fixed Income, Index, Currency, Municipal, Equity and Mortgage securities. An exception to this is Equity Options. The Unique ID is derived from other security-level data for Equity Options.

- ID\_BB\_UNIQUE is unique to a specific security and not the company. If there is more than one ticker for a company within a country, each will have its own ID\_BB\_UNIQUE. If there is more than one country listing, each will have its own ID\_BB\_UNIQUE.
- ID\_BB\_UNIQUE should be used to track a change to a security's Cusip identifier, since such an action occurs often, especially for Collateralized Mortgage Obligations.
- The combination of ID\_BB\_UNIQUE and ID\_BB\_COMPANY should be used to identify an individual security for Mortgages, Munis, and Indices.

**ID\_BB\_COMPANY** – The Bloomberg Company ID can be used to link or join securities of the same company across files. The combination of ID\_BB\_COMPANY and ID\_BB\_SECURITY identifies an individual security for Munis, Equities, Corporates, Governments and Preferreds. The combination of ID\_BB\_COMPANY and ID\_BB\_UNIQUE identifies an individual security for Mortgages and Indices.

**ID\_BB\_SECURITY** – The Bloomberg Security ID should be used in combination with ID\_BB\_COMPANY in order to distinguish between securities of the same company for Munis, Equities, Corporates, Governments and Preferreds. ID\_BB\_SECURITY is a randomly assigned number.

By using these identifiers, the user is able to link all securities with the same issuer or company name whether it is an equity or debt instrument.

## Production Support and Technical Assistance

### North America

| Support Type                                          | Contact Information                                                      |
|-------------------------------------------------------|--------------------------------------------------------------------------|
| Connection/FTP, File Processing, Software, Encryption | 1-212-318-2000                                                           |
| Data Content                                          | <a href="mailto:datalicense@bloomberg.net">datalicense@bloomberg.net</a> |

### Europe, Africa, Middle East

| Support Type                                          | Contact Information                                                      |
|-------------------------------------------------------|--------------------------------------------------------------------------|
| Connection/FTP, File Processing, Software, Encryption | +44-20-7073-3330                                                         |
| Data Content                                          | <a href="mailto:datalicense@bloomberg.net">datalicense@bloomberg.net</a> |

### Mexico, Central & South America

| Support Type                                          | Contact Information                                                      |
|-------------------------------------------------------|--------------------------------------------------------------------------|
| Connection/FTP, File Processing, Software, Encryption | +1-212-318-2000                                                          |
| Data Content                                          | <a href="mailto:datalicense@bloomberg.net">datalicense@bloomberg.net</a> |

### Australia

| Support Type                                          | Contact Information                                                      |
|-------------------------------------------------------|--------------------------------------------------------------------------|
| Connection/FTP, File Processing, Software, Encryption | +612-9777-1243                                                           |
| Data Content                                          | <a href="mailto:datalicense@bloomberg.net">datalicense@bloomberg.net</a> |

### Japan

| Support Type                                          | Contact Information                                                      |
|-------------------------------------------------------|--------------------------------------------------------------------------|
| Connection/FTP, File Processing, Software, Encryption | +81-3-3201-8989                                                          |
| Data Content                                          | <a href="mailto:datalicense@bloomberg.net">datalicense@bloomberg.net</a> |

### Singapore

| Support Type                                          | Contact Information                                                      |
|-------------------------------------------------------|--------------------------------------------------------------------------|
| Connection/FTP, File Processing, Software, Encryption | +65-6212-1492                                                            |
| Data Content                                          | <a href="mailto:datalicense@bloomberg.net">datalicense@bloomberg.net</a> |

### Hong Kong

| Support Type                                          | Contact Information                                                      |
|-------------------------------------------------------|--------------------------------------------------------------------------|
| Connection/FTP, File Processing, Software, Encryption | +852-2977-6130                                                           |
| Data Content                                          | <a href="mailto:datalicense@bloomberg.net">datalicense@bloomberg.net</a> |

## FTP

A file is available on the primary and backup FTP servers for seven (7) days after the initial request. Clients will need to maintain a backup if there is a need to store a file longer.

When a reply file is transferred from our backend server, a copy of it is kept, appended with the date it arrived. For example, a reply file named **prices.out** would be copied to **prices.out.20050501**, where 20050501 represents May 1, 2005. If the reply file **prices.out.20050501** already existed on the FTP server, due to a previous request on the same day, the new reply file would be copied to **prices.out.20050501.1**. If the “.1” version existed, the new reply file would be copied to **prices.out.20050501.2**, and so on. The most recent file would always be stored as **prices.out**.

The correct way to utilize Data License internet FTP servers is to pull the date-stamped file. Internet servers have a failure recovery round robin system that consists of multiple physical machines, pointed to by [bfmrr.bloomberg.com](http://bfmrr.bloomberg.com). Clients who retrieve non-date-stamped output file run the risk of using a stale .out file. This will happen if the client drops a request file on machine1, drops connection, then reconnects to machine2; machine2 may have an older copy of the .out file with the same name.

Each FTP server has a backup server. In the event that the primary server is unavailable, requests can be placed on the backup server. All reply files are placed on the primary and backup servers so that output can be picked up on either server. Files are kept on the backup server and the primary server for seven days. The file backup mechanism is also used for reply files scheduled via FTP.

Please refer to [Appendix B](#) for the list of host addresses.

## Data Support

The data license product support team can be reached at [datalicense@bloomberg.net](mailto:datalicense@bloomberg.net). **The group that is dedicated to monitoring this email address handles all questions and/or concerns regarding data content only; questions of a technical nature need to be addressed directly by the above support desks.** Clients are asked to provide their data license account number and applicable file names with their queries.

## When Changes Are Made

### Changes, Requests, and New Features

#### A. Enhancement Notices

Bloomberg will periodically need to change or update specific elements of the Data License product. Some examples of possible changes/updates include:

##### General Notices

- Adjustments or additions of exchange codes
- Adjustments or additions of corporate action fields
- Adjustments of field types (character, integer, etc.) and/or field widths
- Update of a field's definition
- Change in logic for an existing field
- Development updates (ex. new/updated products, files)
- Additional corporate action types for getactions program
- Additions to output files from getticks, getallticks, and gethistory requests

**Enhancement notices are placed in the Notices folder/subdirectory located in the home directory.**

- These notices will be created as Microsoft Word documents; therefore, clients will need to be in BINARY MODE when retrieving them.
- Clients should check for and receive these notices daily.
- These notices are IMPORTANT! They may require minor programming changes. Please review them carefully in order to prevent production errors.

***For clients with the Bloomberg PROFESSIONAL™ Service, enhancement notices and client software/applications can be downloaded via DLSD<GO>.***

Each notice will contain the following standard information:

- Enhancement notice date
- Short description of the enhancement
- Effective date, if known
- An indication as to whether the notice applies to Per Security and/or Back Office Clients

In addition, these notices will provide a detailed explanation as to what changes will be taking place, and how they should be handled.



## **Naming Convention for Notices Directory Files**

Enhancement and alert notices in the FTP notices subdirectories are currently provided in the following naming scheme:

```
alert_<date in YYYYMMDD format>.doc  
enhance_<date in YYYYMMDD format>.doc  
jp_alert_<date in YYYYMMDD format>.doc  
jp_enhance_<date in YYYYMMDD format>.doc
```

Because Back Office and Per Security information are frequently sent in separate notices, the following naming conventions will be introduced to assist clients in identifying whether a notice involves their product:

### **BACK OFFICE ONLY NOTICES**

```
alert_bo_<date in YYYYMMDD format>.doc  
enhance_bo_<date in YYYYMMDD format>.doc  
jp_alert_bo_<date in YYYYMMDD format>.doc  
jp_enhance_bo_<date in YYYYMMDD format>.doc
```

### **PER SECURITY ONLY NOTICES**

```
alert_ps_<date in YYYYMMDD format>.doc  
enhance_ps_<date in YYYYMMDD format>.doc  
jp_alert_ps_<date in YYYYMMDD format>.doc  
jp_enhance_ps_<date in YYYYMMDD format>.doc
```

### **BOTH BACK OFFICE AND PER SECURITY RELATED NOTICE**

```
alert_all_<date in YYYYMMDD format>.doc  
enhance_all_<date in YYYYMMDD format>.doc  
jp_alert_all_<date in YYYYMMDD format>.doc  
jp_enhance_all_<date in YYYYMMDD format>.doc
```

NOTE: At the end of a calendar year, all enhancement notices for that year will be placed in documents called enhyyyy.doc (English) and enhyyyyj.doc (Japanese) and sent to FTP Notices directories and the DLSD <go> function. For example, the notices from 2004 are available as enh2004.doc and enh2004j.doc

## **B. File Layout Changes**

The fields.csv file is subject to change, as new fields are created and added to the file, definitions are updated, etc. Corporate actions are also subject to change; in the event that new actions are introduced or field elements are added to existing actions, clients will be notified via enhancement notice.

### **C. Field Data Type and Value Changes**

The data/value of an existing data field may be changed if a correction in logic is required. If this is not the case, a new field will be created. An announcement will not be made for mnemonic changes.

### **D. Version Control**

The Data License FTP Client and Request Builder products have been merged into one application; this requires the user to log in only once to use both products. The current version of the software is 4.0. Version numbers are periodically updated, and the new version will replace the old version when posted to the FTP Notices directory or the **DLSD <go>** function on the Bloomberg terminal.

# Appendices

## APPENDIX A. Copyright for Encryption Software

Copyright (C) 1995-1997 Eric Young (eay@mincom.oz.au)  
All rights reserved.

The encryption package is a DES implementation written by Eric Young (eay@mincom.oz.au).  
The implementation was written so as to conform to MIT's libdes.

This library is free for commercial and non-commercial use as long as the following conditions are adhered to. The following conditions apply to all code found in the distribution.

Copyright remains Eric Young's, and as such any Copyright notices in the code are not to be removed. If this package is used in a product, Eric Young should be given attribution as the author of that the SSL library. This can be in the form of a textual message at program startup or in documentation (online or textual) provided with the package.

Redistribution and use in source and binary forms, with or without modification, are permitted provided that the following conditions are met:

1. Redistributions of source code must retain the copyright notice, this list of conditions and the following disclaimer.
2. Redistributions in binary form must reproduce the above copyright notice, this list of conditions and the following disclaimer in the documentation and/or other materials provided with the distribution.
3. All advertising materials mentioning features or use of this software must display the following acknowledgment: This product includes software developed by Eric Young (eay@mincom.oz.au).

THIS SOFTWARE IS PROVIDED BY ERIC YOUNG ``AS IS" AND ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE ARE DISCLAIMED. IN NO EVENT SHALL THE AUTHOR OR CONTRIBUTORS BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION). HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

The license and distribution terms for any publicly available version or derivative of this code cannot be changed. i.e. this code cannot simply be copied and put under another distribution license (including the GNU Public License).

## APPENDIX B. IP Addresses

### Current Dedicated FTP Connections (to be discontinued October 3, 2005):

#### North America/Tokyo:

|     |             |
|-----|-------------|
| FW1 | 160.43.94.2 |
| FW2 | 160.43.94.3 |

#### London

|       |              |
|-------|--------------|
| LFTP1 | 160.43.13.10 |
| LFTP2 | 160.43.13.13 |

### New Dedicated FTP Connections (Effective October 3, 2005)

#### North American/Tokyo

|        |               |
|--------|---------------|
| ftp2ny | 160.43.94.20  |
| ftp2nj | 160.43.166.57 |

#### London

|        |               |
|--------|---------------|
| lftpny | 160.43.94.24  |
| lftpnj | 160.43.166.58 |

### All Regions      Internet FTP Connection:

BFMRR:

Host Name:    bfmrr.bloomberg.com

Clients are advised to use **bfmrr.bloomberg.com** for all interaction with Bloomberg internet servers. This is a round robin DNS that alternates between two servers. In the event that one of these two machines becomes unavailable, bfmrr.bloomberg.com will point only to the live machine.

## APPENDIX C. Sample Files for Customizing Output Format

### Request file

```
START-OF-FILE
SYSTEM=DATA
LOGIN=JDOE
USERNUMBER=111111
REPLYFILENAME=Sample1.txt
PROGRAMNAME=getdata
PROGRAMFLAG =oneshot
SECMASTER=yes
SN=777777
WS=0
LAUNCH=YES
OUTPUTFORMAT=variable
SECDESLength=14
```

```
START-OF-FIELDS
NAME
NAME|16
PX_LAST
PX_LAST|14
PX_LAST|14|4
PX_LAST|14|4|c
PX_LAST||8
PX_LAST|||c
END-OF-FIELDS
```

```
START-OF-DATA
IBM US Equity||
END-OF-DATA
```

```
END-OF-FILE
```

## Output file

START-OF-FILE  
SYSTEM=DATA  
LOGIN=JDOE  
USERNUMBER=111111  
REPLYFILENAME=Sample1.txt  
PROGRAMNAME=getdata  
PROGRAMFLAG =oneshot  
SECMASTER=yes  
SN=777777  
WS=0  
LAUNCH=YES  
OUTPUTFORMAT=variable  
SECDESLength=14

START-OF-FIELDS  
NAME  
NAME|16  
PX\_LAST  
PX\_LAST|14  
PX\_LAST|14|4  
PX\_LAST|14|4|c  
PX\_LAST||8  
PX\_LAST|||c  
END-OF-FIELDS

TIMESTARTED=Thu Oct 16 18:12:46 EDT 1997  
START-OF-DATA  
IBM US Equity |0|8|INTL BUSINESS MACHINES CORP|INTL BUSINESS  
MA|99.875000|  
99.875000 |99.8750 |4 998750|99.87500000|699875000|  
END-OF-DATA  
TIMEFINISHED=Thu Oct 16 18:12:58 EDT 1997  
END-OF-FILE

## **Request file**

```
START-OF-FILE
SYSTEM=DATA
LOGIN=JDOE
USERNUMBER=111111
REPLYFILENAME=Sample2.txt
PROGRAMNAME=getdata
PROGRAMFLAG =oneshot
SECMASTER=yes
SN=777777
WS=0
LAUNCH=YES
OUTPUTFORMAT=variable
SECDESLength=14
DELIMITER=;
```

```
START-OF-FIELDS
NAME
NAME|16
PX_LAST
PX_LAST|14
PX_LAST|14|4|c
PX_LAST||8
PX_LAST|||c
END-OF-FIELDS
```

```
START-OF-DATA
IBM US Equity|
END-OF-DATA
```

```
END-OF-FILE
```

## Output file

```
START-OF-FILE
SYSTEM=DATA
LOGIN=JDOE
USERNUMBER=111111
REPLYFILENAME=Sample2.txt
PROGRAMNAME=getdata
PROGRAMFLAG =oneshot
SECMaster=yes
SN=777777
WS=0
LAUNCH=YES
OUTPUTFORMAT=variable
SECDESLength=14
DELIMITER=;
```

```
START-OF-FIELDS
NAME
NAME|16
PX_LAST
PX_LAST|14
PX_LAST|14|4|c
PX_LAST||8
PX_LAST|||c
END-OF-FIELDS
```

```
TIMESTARTED=Thu Oct 16 18:17:36 EDT 1997
START-OF-DATA
"IBM US Equit";0;7;"INTL BUSINESS MACHINES CORP";"INTL BUSINESS
";99.875000 ;99.875000 ;4 998750;99.87500000 ;6 99875000;
END-OF-DATA
TIMEFINISHED=Thu Oct 16 18:17:48 EDT 1997
END-OF-FILE
```



## **Request file**

```
START-OF-FILE  
SYSTEM=DATA  
LOGIN=JDOE  
USERNUMBER=111111  
REPLYFILENAME=Sample3.txt  
PROGRAMNAME=getdata  
PROGRAMFLAG =oneshot  
SECMASTER=yes  
SN=777777  
WS=0  
LAUNCH=YES  
OUTPUTFORMAT=fixed  
SECDESLength=12  
DELIMITER=;
```

```
START-OF-FIELDS  
NAME  
NAME|16  
PX_LAST  
PX_LAST|14|4|c  
PX_LAST||8  
PX_LAST|||c  
END-OF-FIELDS
```

```
START-OF-DATA  
IBM US Equity||  
END-OF-DATA
```

```
END-OF-FILE
```

## Output file

START-OF-FILE  
SYSTEM=DATA  
LOGIN=JDOE  
USERNUMBER=111111  
REPLYFILENAME=Sample3.txt  
PROGRAMNAME=getdata  
PROGRAMFLAG =oneshot  
SECMASTER=yes  
SN=777777  
WS=0  
LAUNCH=YES  
OUTPUTFORMAT=fixed  
SECDESLength=12  
DELIMITER=;

START-OF-FIELDS  
NAME  
NAME|16  
PX\_LAST  
PX\_LAST|14|4|c  
PX\_LAST||8  
PX\_LAST|||c  
END-OF-FIELDS

TIMESTARTED=Thu Oct 16 18:20:26 EDT 1997  
START-OF-DATA  
IBM US Equit0 6 INTL BUSINESS MACHINES CORP INTL BUSINESS  
MA99.875000 4 99875099.87500006 99875000  
END-OF-DATA  
TIMEFINISHED=Thu Oct 16 18:20:38 EDT 1997  
END-OF-FILE

## APPENDIX D. Lookup Tables

Lookup tables are special fields that may be found in the Data Dictionary (fields.csv); please see Data Dictionary Section. All the Lookup tables begin with “LU\_”. These return all possible values for the fields that they are associated with, for the specified market sector (yellow key). For example, **LU\_SECURITY\_TYP** requested for an equity will return all possible equity values for the field **SECURITY\_TYP**.

This information is returned in bulk format. When requesting a lookup table, any security may be used that is recognizable, even though the security may not belong to the same market sector as the information returned by the lookup table. (i.e., “IBM US Equity” may be used to obtain every lookup table field in fields.csv).

For example, the following request file will return all possible values for MARKET\_SECTOR\_DES:

```
START-OF-FILE
FIRMNAME=dlsample
REPLYFILENAME=sample.dat
PROGRAMNAME=getdata
PROGRAMFLAG=one-shot
```

```
START-OF-FIELDS
LU_MARKET_SECTOR
END-OF-FIELDS
```

```
START-OF-DATA
IBM US Equity
END-OF-DATA
END-OF-FILE
```

The corresponding output would be:

```
START-OF-FILE
FIRMNAME=dlsample
REPLYFILENAME=sample.dat
PROGRAMNAME=getdata
PROGRAMFLAG=one-shot
```

```
START-OF-FIELDS
LU_SECURITY_TYP
END-OF-FIELDS
```

```
TIMESTARTED=Thu Jan 13 13:41:32 EST 2005
START-OF-DATA
IBM US Equity|0|1|;2;58;1;1;A/T Prime;1;A/T Score;1;A/T
Unit;1;ADR;1;Amertst;1;Austrian Crt;1;BDR;1;Basket WRT;1;Belgium
Cert;1;Bond;1;CDR;1;Car Forward;1;Closed-End Fund;1;Cmndt Fut
WRT;1;Cmndt Idx WRT;1;Common Stock;1;Conv Bond;1;Conv Prfd;1;Corp
Bnd WRT;1;Currency WRT;1;Dutch Cert;1;EDR;1;ETF;1;Equity
Option;1;Equity WRT;1;French
```

Cert;1;GDR;1;German Cert;1;Hedge Fund;1;I.R. Fut WRT;1;I.R. Swp  
WRT;1;IDR;1;Index WRT;1;Indx Fut WRT;1;Int. Rt. WRT;1;Ltd  
Part;1;Misc.;1;NY Reg Shrs;1;OTC Option;1;Open-End  
Fund;1;PRES;1;Preference;1;Preferred;1;Prfd WRT;1;Private  
Comp;1;Private Eqty;1;RDC;1;REIT;1;Receipt;1;Right;1;Royalty  
Trst;1;Swiss Cert;1;Tracking Stk;1;Trusts;1;UIT;1;Unit;1;Unit Inv  
Tst;1;Warrant;|  
END-OF-DATA  
TIMEFINISHED=Thu Jan 13 13:41:34 EST 2005  
END-OF-FILE

## APPENDIX E. Bloomberg Special Fonts

It is possible that data retrieved from Bloomberg contains special fonts. All of the fonts are shown below:

### BLOOMBERG Font

|    | _0                            | _1                            | _2                            | _3                            | _4                            | _5                            | _6                            | _7                          | _8                            | _9                            | _A                            | _B                            | _C                            | _D                            | _E                            | _F                          |
|----|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-----------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-----------------------------|
| 0_ | Ç                             | ü                             | é                             | â                             | ä                             | à                             | ã                             | ç                           | ê                             | ë                             | è                             | ï                             | ì                             | í                             | î                             | ÿ                           |
| 1_ | É                             | È                             | Ì                             | ô                             | ö                             | ò                             | û                             | ù                           | ÿ                             | Ö                             | Ü                             | á                             | í                             | ó                             | ú                             | ñ                           |
| 2_ |                               | !                             | "                             | #                             | \$                            | %                             | &                             | '                           | (                             | )                             | *                             | +                             | ,                             | -                             | .                             | /                           |
| 3_ | 0                             | 1                             | 2                             | 3                             | 4                             | 5                             | 6                             | 7                           | 8                             | 9                             | :                             | ;                             | <                             | =                             | >                             | ?                           |
| 4_ | @                             | A                             | B                             | C                             | D                             | E                             | F                             | G                           | H                             | I                             | J                             | K                             | L                             | M                             | N                             | O                           |
| 5_ | P                             | Q                             | R                             | S                             | T                             | U                             | V                             | W                           | X                             | Y                             | Z                             | [                             | \                             | ]                             | ^                             | _                           |
| 6_ | '                             | a                             | b                             | c                             | d                             | e                             | f                             | g                           | h                             | i                             | j                             | k                             | l                             | m                             | n                             | o                           |
| 7_ | p                             | q                             | r                             | s                             | t                             | u                             | v                             | w                           | x                             | y                             | z                             | {                             |                               | }                             | ~                             | €                           |
| 8_ | <sup>1</sup> / <sub>64</sub>  | <sup>1</sup> / <sub>32</sub>  | <sup>3</sup> / <sub>64</sub>  | <sup>1</sup> / <sub>16</sub>  | <sup>5</sup> / <sub>64</sub>  | <sup>3</sup> / <sub>32</sub>  | <sup>7</sup> / <sub>64</sub>  | <sup>1</sup> / <sub>8</sub> | <sup>9</sup> / <sub>64</sub>  | <sup>5</sup> / <sub>32</sub>  | <sup>11</sup> / <sub>64</sub> | <sup>3</sup> / <sub>16</sub>  | <sup>13</sup> / <sub>64</sub> | <sup>7</sup> / <sub>32</sub>  | <sup>15</sup> / <sub>64</sub> | <sup>1</sup> / <sub>4</sub> |
| 9_ | <sup>17</sup> / <sub>64</sub> | <sup>3</sup> / <sub>32</sub>  | <sup>19</sup> / <sub>64</sub> | <sup>5</sup> / <sub>16</sub>  | <sup>21</sup> / <sub>64</sub> | <sup>11</sup> / <sub>32</sub> | <sup>23</sup> / <sub>64</sub> | <sup>3</sup> / <sub>8</sub> | <sup>25</sup> / <sub>64</sub> | <sup>13</sup> / <sub>32</sub> | <sup>27</sup> / <sub>64</sub> | <sup>17</sup> / <sub>16</sub> | <sup>29</sup> / <sub>64</sub> | <sup>15</sup> / <sub>32</sub> | <sup>31</sup> / <sub>64</sub> | <sup>1</sup> / <sub>2</sub> |
| A_ | <sup>33</sup> / <sub>64</sub> | <sup>17</sup> / <sub>32</sub> | <sup>35</sup> / <sub>64</sub> | <sup>9</sup> / <sub>16</sub>  | <sup>37</sup> / <sub>64</sub> | <sup>19</sup> / <sub>32</sub> | <sup>39</sup> / <sub>64</sub> | <sup>5</sup> / <sub>8</sub> | <sup>41</sup> / <sub>64</sub> | <sup>21</sup> / <sub>32</sub> | <sup>43</sup> / <sub>64</sub> | <sup>11</sup> / <sub>16</sub> | <sup>45</sup> / <sub>64</sub> | <sup>23</sup> / <sub>32</sub> | <sup>47</sup> / <sub>64</sub> | <sup>3</sup> / <sub>4</sub> |
| B_ | <sup>49</sup> / <sub>64</sub> | <sup>25</sup> / <sub>32</sub> | <sup>51</sup> / <sub>64</sub> | <sup>13</sup> / <sub>16</sub> | <sup>53</sup> / <sub>64</sub> | <sup>27</sup> / <sub>32</sub> | <sup>55</sup> / <sub>64</sub> | <sup>7</sup> / <sub>8</sub> | <sup>57</sup> / <sub>64</sub> | <sup>29</sup> / <sub>32</sub> | <sup>59</sup> / <sub>64</sub> | <sup>15</sup> / <sub>16</sub> | <sup>61</sup> / <sub>64</sub> | <sup>31</sup> / <sub>32</sub> | <sup>63</sup> / <sub>64</sub> | ×                           |
| C_ | 0)                            | 1)                            | 2)                            | 3)                            | 4)                            | 5)                            | 6)                            | 7)                          | 8)                            | 9)                            | 0                             | 1                             | 2                             | 3                             | 4                             | 5                           |
| D_ | 6                             | 7                             | 8                             | 9                             | ↑                             | ↓                             | ←                             | →                           | ↗                             | ↘                             | ↖                             | ↙                             | ¼                             | ½                             | ¾                             | ⅞                           |
| E_ | £                             | ¥                             | fr                            | ð                             | ù                             | ±                             | ≠                             | ≈                           | ≤                             | ≥                             | ö                             | á                             | í                             | ™                             | ©                             | ®                           |
| F_ | ô                             | ✓                             | ó                             | ú                             | â                             | ê                             | õ                             | à                           | ñ                             | ¿                             | ¡                             | «                             | »                             | ã                             | ä                             | ß                           |

Since many of these characters are not recognized as standard ASCII, a conversion is done so that all characters are in ASCII form. The table on the following page shows how the Bloomberg special characters are mapped except for those characters which have the same ASCII translation. Fractions are not shown in the next table but are handled in the following manner: <space><numerator>/<denominator>. The additional space was used to avoid potential confusion with numbers that may be immediately before the fraction.

| Special Char | ASCII Map | Special Char | ASCII Map | Special Char   | ASCII Map | Special Char | ASCII Map | Special Char | ASCII Map |
|--------------|-----------|--------------|-----------|----------------|-----------|--------------|-----------|--------------|-----------|
| Ç            | C         | É            | E         | €              | =         | Ò            | O         | Ú            | U         |
| ü            | u         | È            | E         | ×              | X         | Û            | U         | Â            | A         |
| é            | e         | Ì            | I         | ↑              | ^         | ±            | +/-       | Ê            | E         |
| â            | a         | ô            | o         | ↓              | v         | ≠            | !=        | õ            | o         |
| ä            | a         | ö            | o         | ←              | <-        | ≈            | ~         | À            | A         |
| à            | a         | ò            | o         | →              | ->        | ≤            | <=        | Ñ            | N         |
| å            | a         | û            | u         | ↗              | />        | ≥            | >=        | ¿            | !?        |
| ç            | c         | ù            | u         | ↙              | </        | Õ            | O         | ¡            | !!        |
| ê            | e         | ÿ            | y         | ↖              | <\        | Á            | A         | «            | <<        |
| ë            | e         | Ö            | O         | ↘              | \>        | Í            | I         | »            | >>        |
| è            | e         | Ü            | U         | W <sub>I</sub> | WI        | ™            | TM        | ã            | a         |
| ï            | i         | á            | a         | P <sub>F</sub> | PF        | ©            | C         | Ã            | A         |
| î            | i         | í            | i         | R <sub>T</sub> | RT        | ®            | R         | ß            | B         |
| ì            | i         | ó            | o         | W <sub>R</sub> | WR        | Ô            | O         |              |           |
| Ä            | A         | ú            | u         | £              | GBP       | √            | V         |              |           |
| Å            | A         | ñ            | n         | ¥              | Yen       | Ó            | O         |              |           |

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